3. Link Layer

3.1. Introduction

3.2. From Bit Streams to Sequences of Frames

3.3. Message Formats in Local Area Networks

3.4. Error Correction

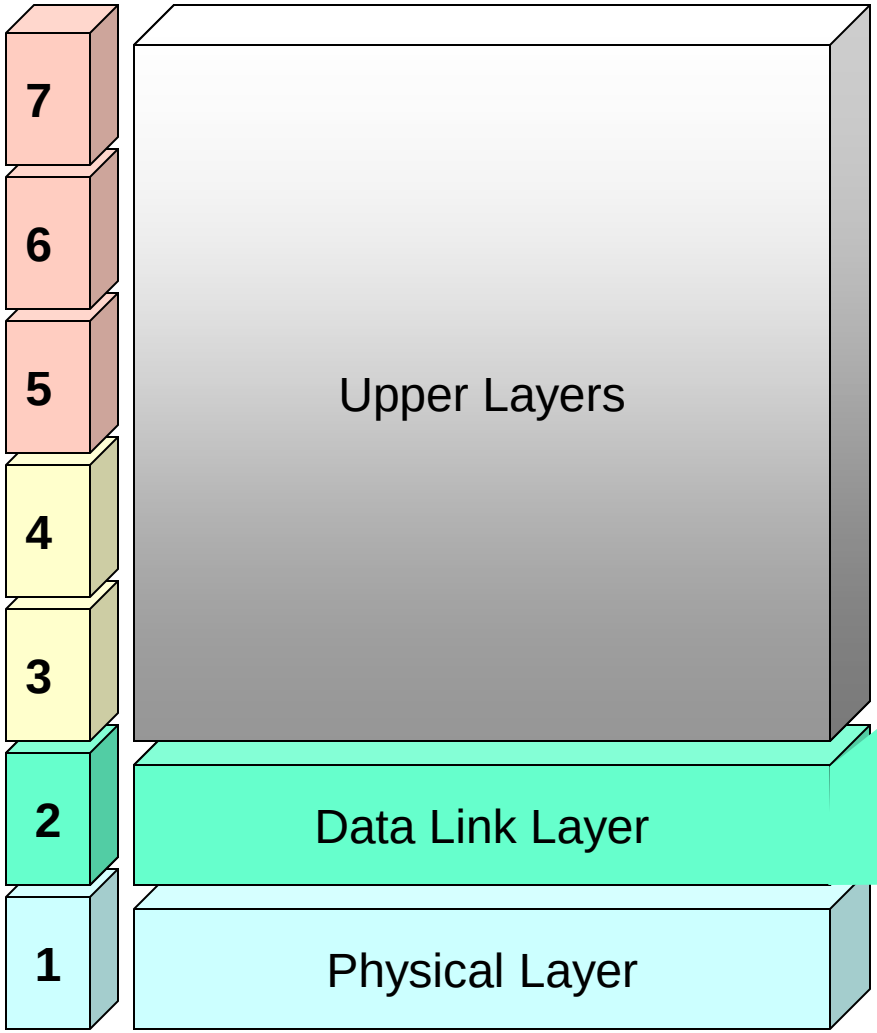
3.5. Multiple Access Schemes

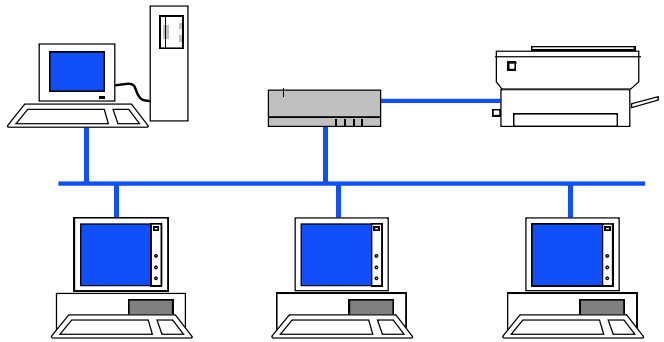
3.6. Medium Access Control

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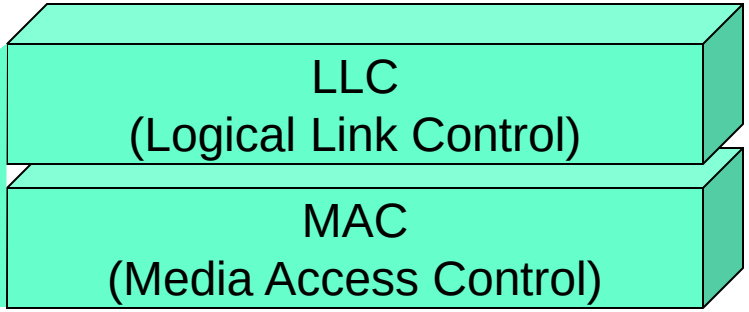
The Layer Model of the IEEE LMSC (“IEEE 802”)

LAN/MAN Standards Committee der IEEE (“IEEE 802”)



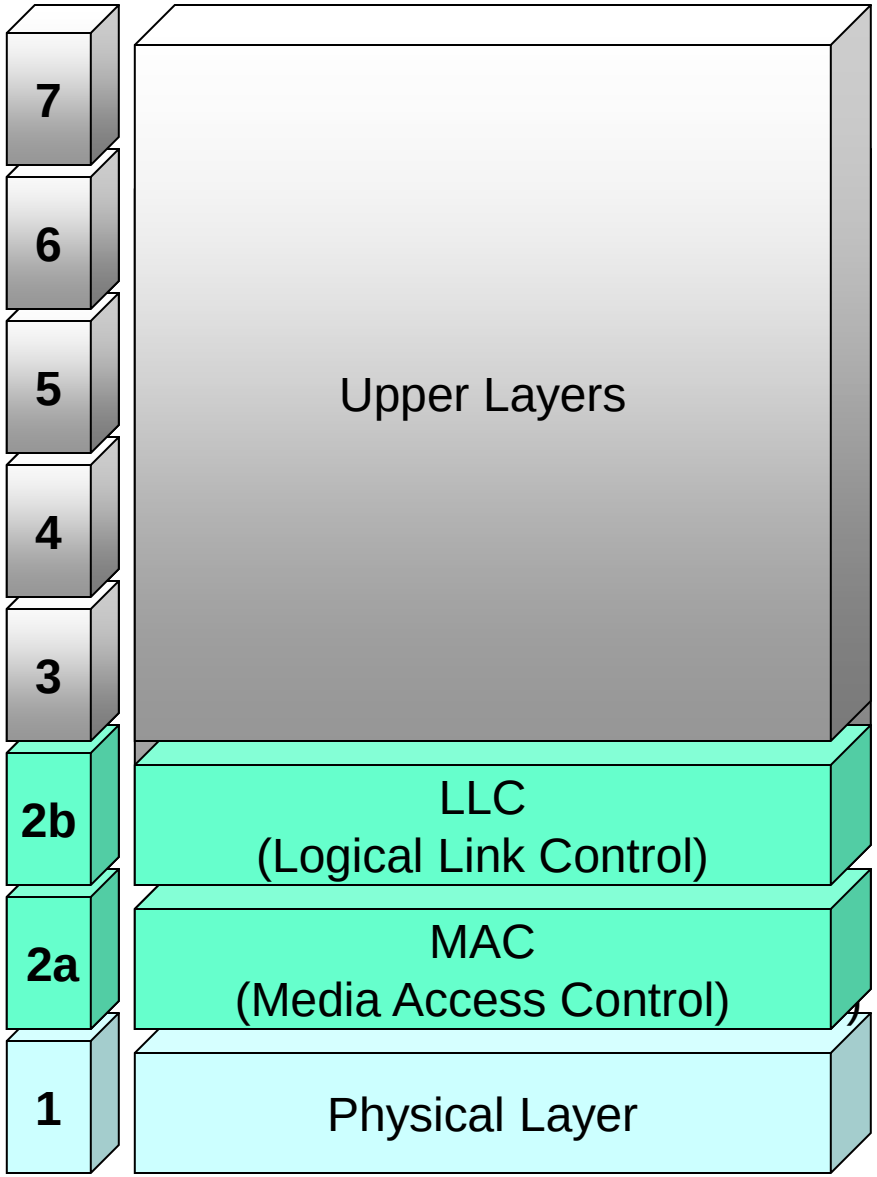


How to control the access to the media?



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The Layer Model of the IEEE LMSC (“IEEE 802”) (2)



The LAN/MAN Standards Committee of the IEEE (better known as “IEEE 802”) standardizes protocols for

- **layer 1,**
- **layer 2a and**
- **layer 2b**

of the OSI model.

Well known protocols are

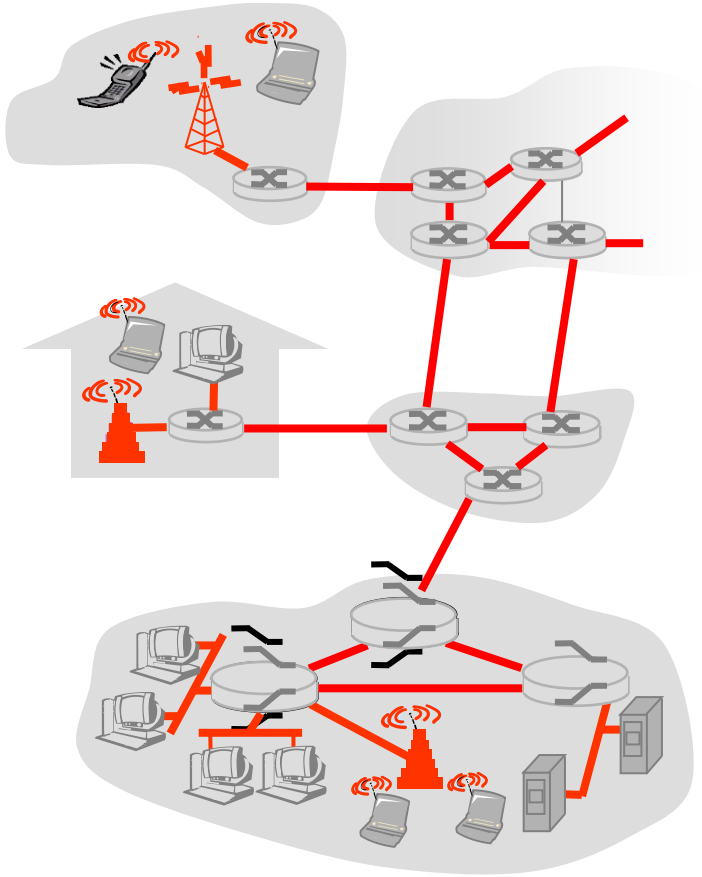
- “**Ethernet**” (IEEE 802.3),
- “**Token Ring**” (IEEE 802.5)
- “**Wireless LAN**” (IEEE 802.11)
- “**Wireless PAN**”, “Bluetooth” (IEEE 802.15)

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3.1 Introduction

Some terminology:

- hosts and routers are **nodes**
- communication channels that connect adjacent nodes along communication path are **links**
  - wired links
  - wireless links
  - LANs
- layer-2 packet is a **frame**, encapsulates datagram



Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

data-link layer has responsibility of transferring datagram from one node to adjacent node over a link

Link layer: context

- **datagram** transferred by **different link protocols** over different links:
  - e.g., Ethernet on first link,
  - frame relay on intermediate links,
  - 802.11 on last link
- each link protocol provides **specific services**

transportation analogy

- trip from Princeton to Lausanne
  - limo: Princeton to JFK
  - plane: JFK to Geneva
  - train: Geneva to Lausanne
- tourist = datagram
- transport segment = communication link
- transportation mode = link layer protocol
- travel agent = routing algorithm

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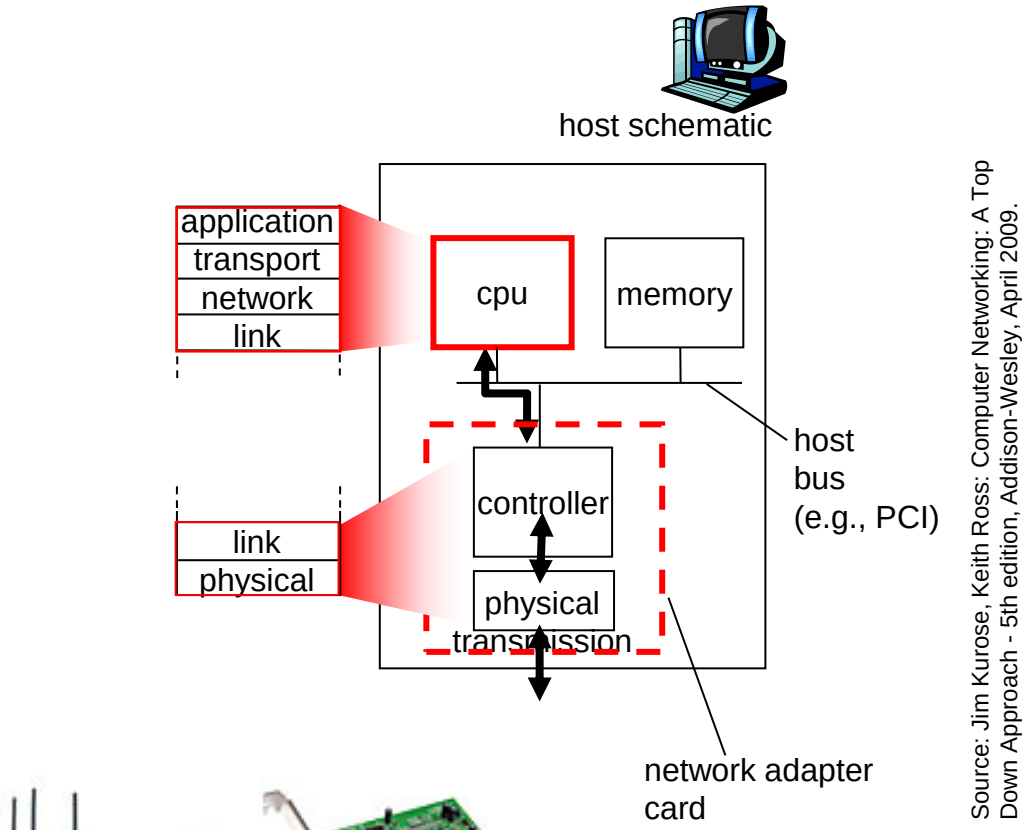
Link layer services

- **framing, link access:**
  - encapsulate datagram to form a frame (adding header trailer)
  - channel access if shared medium
  - “MAC” addresses used in frame headers to identify source, destination
    - different from IP address!
- **reliable delivery between adjacent nodes**
  - rarely used on low bit-error link
    - (fiber, some twisted pair)
  - wireless links: high error rates
- **flow control:**
  - pacing between adjacent sending and receiving nodes
- **error detection:**
  - errors caused by signal attenuation, distortion, noise.
  - receiver detects presence of errors:
    - asks sender to retransmit or simply drops frame
- **error correction:**
  - receiver locates and corrects bit error(s) without resorting to retransmission

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Link layer: Where is it implemented?

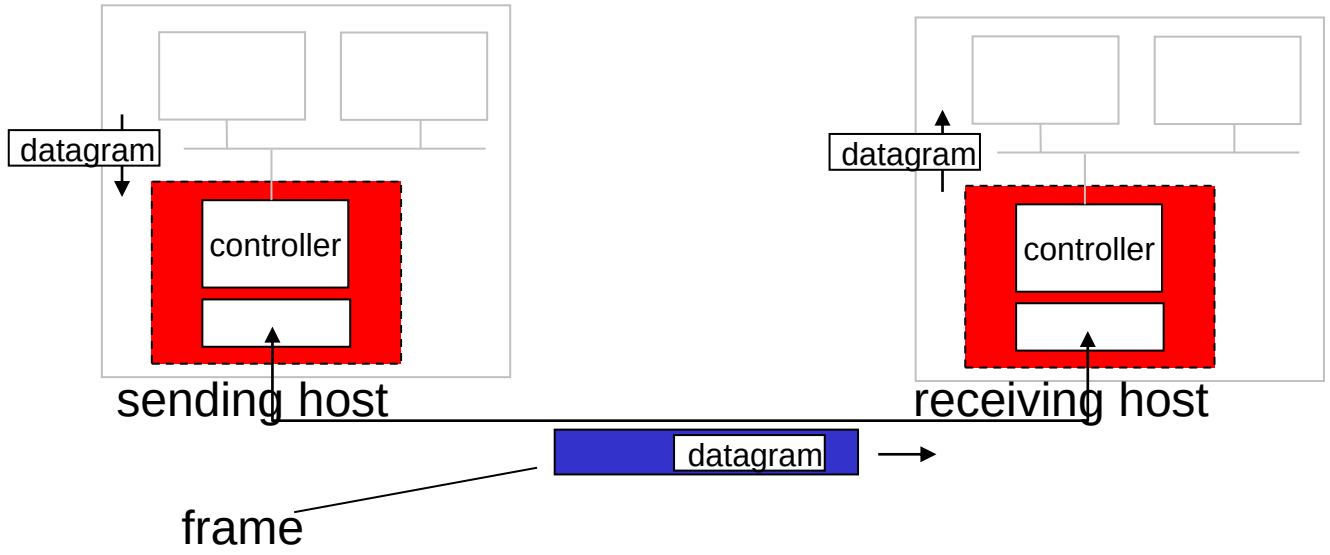
- in each and every host
- link layer implemented in “adaptor” (aka **network interface card** NIC)
  - Ethernet card, PCMCI card, 802.11 card
  - implements link, physical layer
- attached to host’s system buses
- combination of hardware, software, firmware





Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

Link layer: Adapters Communicating



Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

- sending side:
  - encapsulates datagram in frame
  - adds error checking bits, flow control, etc.
- receiving side
  - checks for errors, flow control, etc
  - extracts datagram, passes to upper layer at receiving side

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3.2. From Bit Streams to Sequences of Frames

Messaging between computers and/or network nodes requires two kinds of synchronization:

a) Bit synchronization (layer 1)

- ☞ Detection of **start and end of individual bits**

b) PDU synchronization (higher layers)

- ☞ Detection of **PDU start and PDU end**
- ☞ Distinction between **“user data” and control information**

The physical layer deals with the transmission and reception of a sequence of (maybe erroneous) bits.

From this bit sequence, the layer 2 PDUs (the “frames”) need to be derived.

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Character-oriented and Length-oriented Protocols

Character-oriented protocols:

In the **60s of the 20th century**, several character-oriented protocols were developed for what we call “layer 2” today. Some of these are still used in legacy systems.

Characteristics of character-oriented protocols:

- A **specific code** is required.
  (Control characters are part of this code. Example: ASCII includes NAK, ACK, ...)
- **Control characters** inside the user data are **“wrapped”**.

Length-oriented protocols:

In the **70s of the 20th century**, several length-oriented protocols were developed: A **length field in the header** tells the **length of the user data field**.

Result:

Control sequences inside the user data field are irrelevant: User data are not accepted as protocol control information.

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Bit-oriented Protocols

Early bit-oriented protocols were also developed in the 70s of the 20th century.
A lot of modern protocols belong to this class.

Characteristics:

- Message transmission in variable length frames with fixed structure.
- **Start and end delimiters** (specific bit strings) mark the beginning and the end of frames.
- Protocol control information is placed at specific bit positions, relative to the start and end delimiters.

The bit-oriented protocol **SDLC** (Synchronous Data Link Control, developed by IBM) was

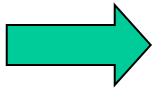
- standardized by ISO in a modified version as **HDLC** (High-level Data Link Control),
- standardized by CCITT (today: ITU) in a modified version as **LAP** (Link Access Protocol).

LAP is an ancestor of both **LAPB and LAPD (ISDN)**.

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Bitstuffing

If the **start / end delimiter** (e.g. 01111110) must not be used inside the PDU **data field** (user data), "bitstuffing" solves the problem:

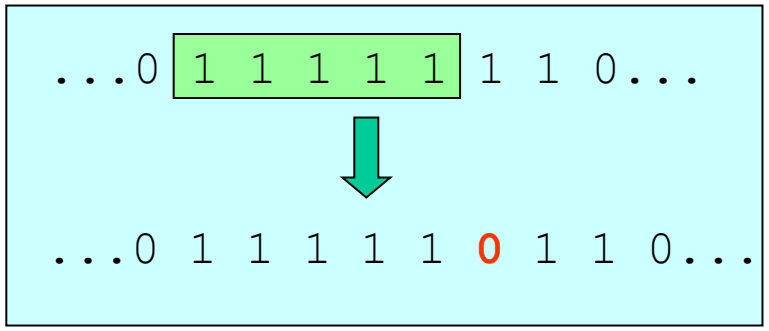


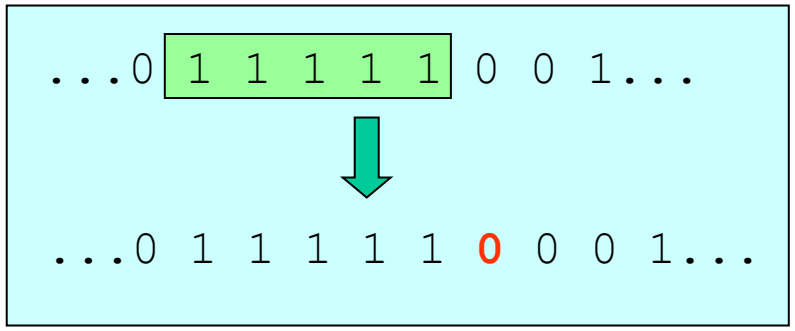
"Insertion of Zeros" (or ones)

Let: `01111110` be the start / end delimiter

Idea: The sender **inserts a "0" after any five "1"s in a row**.

The **receiver eliminates this "0"** before forwarding the data to the user.





Note: Erroneous bits may result in **"Gained Zeros"** !

Let ...0 1 1 1 1 1 0 1 1...

be changed to ...0 1 0 1 1 1 0 1 1...,

In this case, the receiver **does not detect the inserted "0"**.

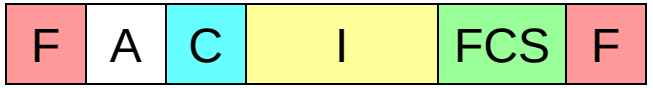
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HDLC: High-Level Data Link Control

HDLC

- was standardized by ISO as ISO 3309 and ISO 4335 as layer 2 protocol,
- **illustrates characteristic structures of bit based protocols**.

Structure of a HDLC frame:



F:	**Start of frame delimiter** (Flag): 01111110	8 bit
A:	**Address field** (historic)	8 bit
C:	**Control field**	8 bit
I :	**User data**	variable
FCS:	**Error detection** (Frame Checking Sequence)	16 bit
F:	**End of frame delimiter** (Flag): 01111110 Uniqueness achieved by “bitstuffing”.	8 bit

Specific meanings were defined for:

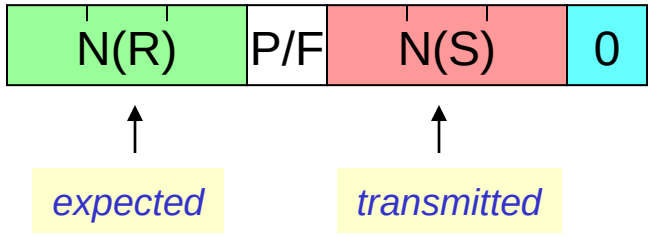
Sequences of ≥ 7 but < 15 ones: signals **line problems**

Sequences of ≥ 15 ones: **Idle** (also used for changing direction in case of half duplex)

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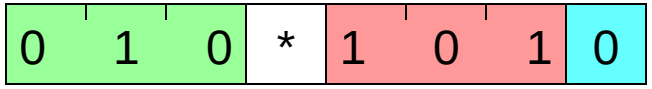
The HDLC control field

HDLC control field structure:



O	=	identifier for information frame (O = 0) or control frame (O = 1)
N(S)	=	Frame number (3 bit)
P/F	=	Poll/Final bit (historic)
N(R)	=	Number of frame expected next *(acks all frames in opposite direction until and including N(R) - 1)*

Example:



Frame 5 is transmitted;

Frame 2 is expected;

ACK until and including frame 1.

How about buffers ?

- **Buffers required at sender** (retransmission of lost frames).
- **No buffer required at receiver in case of "Go-back-N"**: All frames transmitted after the lost or erroneous frame are retransmitted. If the option "Selective Reject" is used, buffers are required at the receiver.

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HDLC frame types

Nr.	Type	Class of frame	C field 8	7	6	5	4	3	2	1
1	I	I (Data frame)	N(R)			P	N(S)			0
2.1	S	RR (Receive Ready)	N(R)			P/F	0	0	0	1
2.2	S	RNR (Receive Not Ready)	N(R)			P/F	0	1	0	1
2.3	S	REJ (Reject)	N(R)			P/F	1	0	0	1
2.4	S	SREJ (Selective Reject)	N(R)			P/F	1	1	0	1
3.1	U	SABM (Set Asynchronous Balanced Mode)	0	0	1	P	1	1	1	1
3.2	U	DISC (Disconnect)	0	1	0	P	0	0	1	1
3.3	U	UA (Unnumbered Acknowledgement)	0	1	1	F	0	0	1	1
3.4	U	CMDR (Command Reject)	1	0	0	F	0	1	1	1
3.5	U	FRMR (Frame Reject)	1	0	0	F	0	1	1	1
3.6	U	DM (Disconnect Mode)	0	0	0	F	1	1	1	1

← Acknowledgement only

← All frames starting from N(R) are rejected

← Only N(R) is rejected

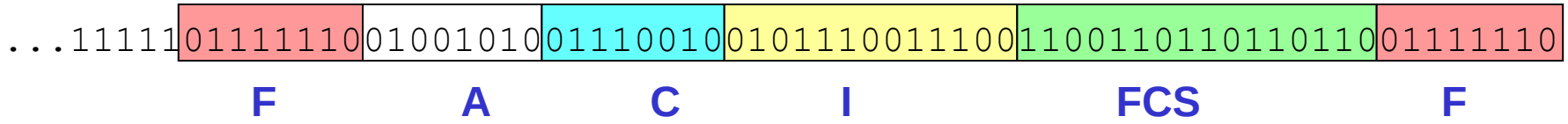
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Example:

The bit string

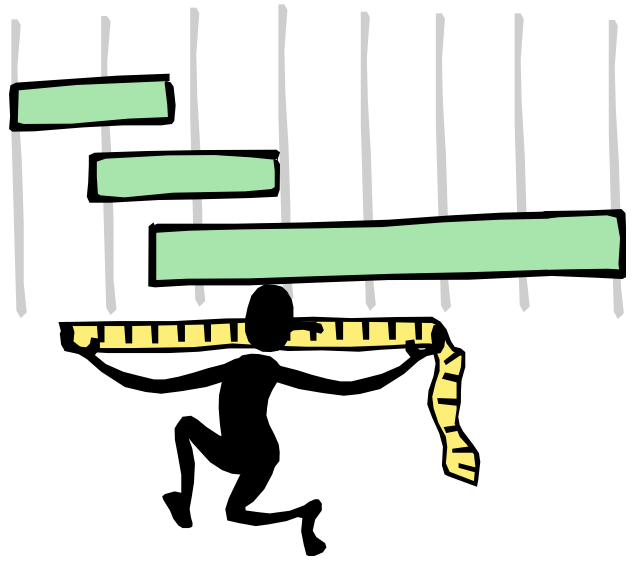
...11111011111100100101001110010010111001110011001101101101100111111 0

is received as:



Message Length

The longer the message, the higher the risk that it is hit by an error on its way to the destination. On the other hand, PCI overhead percentage is smaller in large PDUs.



The **data field** (user data) should be

- **large**,
  - to reduce overall PCI overhead,
  - to efficiently use high bandwidth channels.
- **short**,
  - if the bit error rate is high,
  - if it takes long to fill the user data field.
    (e.g. “multimedia communication”)

We may use **fixed** or **variable length** frames.

In case of variable length, **lower** and/or **upper bounds** may be used.

Fixed length frames simplify the implementation considerably.
However, the **choice** of an “appropriate” length turns out to be **extremely difficult**.

3.3 Message Formats in Local Area Networks

In conventional Local Area Networks, LANs,

- the stations **share a common medium**,
- **all** stations **“see” all messages** in “their” LAN,
- the **receiver is identified by the destination address** (layer 2 address),
- **messages** usually have **variable length**.

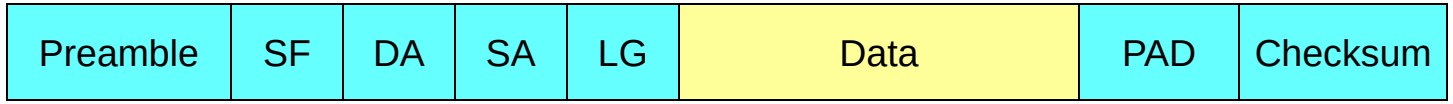
3.3.1. IEEE 802.3 Frame Format

3.3.2. Ethernet Frame Format

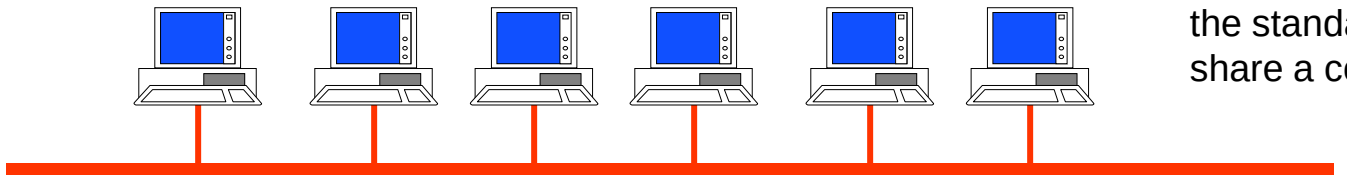
3.3.3. Wireless LAN (IEEE 802.11) Frame Format

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3.3.1. IEEE 802.3 Frame Format



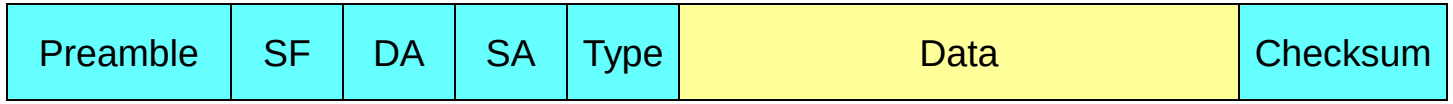
Preamble:	7 byte	**Synchronization**
SF:	1 byte	**Start of frame** delimiter
DA:	6 byte	**Destination** address (MAC address of receiver)
SA:	6 byte	**Source** address (MAC address of sender)
LG:	2 byte	**Length** of data field
Data:	0 - 1500 byte	User **data** (e.g. IP datagram)
PAD:	0-46 byte	**Pad bits** fill up the frame to the minimum size
Checksum:	4 byte	**Cyclic redundancy check** for error detection



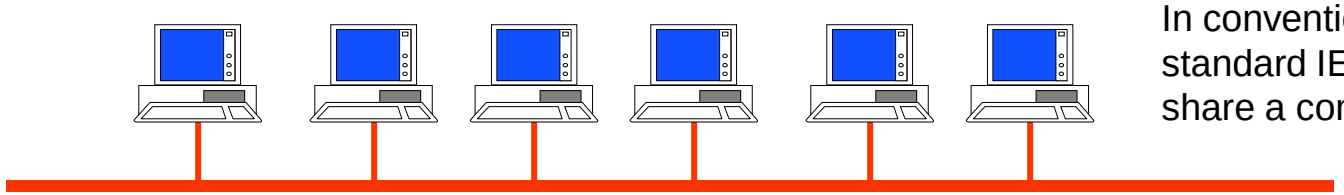
In conventional LANs according to the standard IEEE 802.3, all stations share a common linear medium.

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3.3.2. Ethernet Frame Format



Preamble:	7 byte	**Synchronization**
SF:	1 byte	**Start of frame** delimiter
DA:	6 byte	**Destination** address (MAC address of receiver)
SA:	6 byte	**Source** address (MAC address of sender)
Type:	2 byte	Identification of **upper layer protocol**, e.g. “IP”
Data:	0 - 1500 byte	User **data** (e.g. IP datagram)
Checksum:	4 byte	**Cyclic redundancy check** for error detection



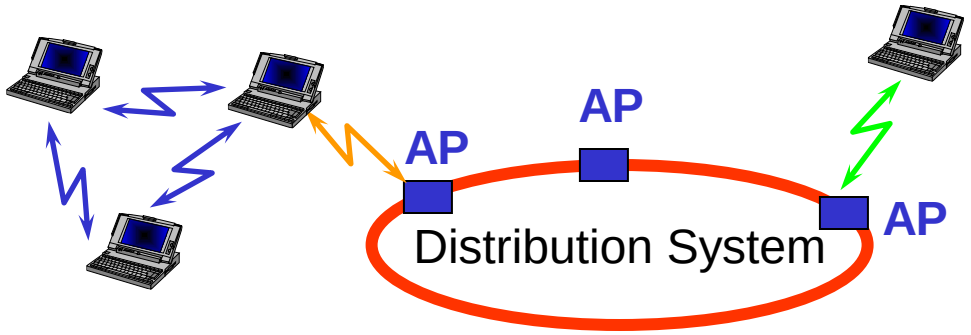
In conventional LANs according to the standard IEEE 802.3, all stations share a common linear medium.

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3.3.3. Wireless LAN (IEEE 802.11) Frame Format

Frame Control	Duration/ ID	Address 1	Address 2	Address 3	Sequence Control	Address 4	Data	FCS

Frame Control:	2 byte	**11 Fields**, e.g. Protocol Version, Type (e.g. Management)
Duration/ID:	2 byte	**Duration** or **"Association Identity"**
Address 1:	6 byte	**Address** (final destination or "basic service set ID" or immediate recipient)
Address 2:	6 byte	**Address** (source address or "basic service set ID" or transmitter address)
Address 3:	6 byte	**Address** (final destination or "basic service set ID" or source address)
Sequence Control:	2 byte	**Sequence number** and **fragment number**
Address 4:	6 byte	**Address** (N/A or source address)
Data:	0 - 2312 byte	User **data** (e.g. IP-Datagram)
FCS:	4 byte	**Frame Check Sequence** for error detection



In Wireless LANs according to the standard IEEE 802.11, the stations either communicate directly or via a "distribution system", DS, with access points, APs.

3.4. Error Correction

There are two possibilities for error correction:

a) Error correction by retransmission

(e.g. in HDLC, LLC, LAPB, TCP, TP4, ...)

→ *Receiver detects error, message is retransmitted*

b) FEC (Forward Error Correction)

→ *Only a subset of bit strings are valid code words. Other bit strings are corrected to the "closest" codeword.*

Def.: Hamming distance

Let $u = (u_{n-1}, u_{n-2}, \ldots, u_0)$ and $v = (v_{n-1}, v_{n-2}, \ldots, v_0)$ be two code words.

Then

$$d(u,v) := \sum_{i=0}^{n-1} |u_i - v_i|$$

is called "Hamming distance" of these words (number of different bits in case of binary codes).

The **minimum Hamming distance of all pairs of (different) code words** is called the **Hamming distance of the code**.

A simple parity check (all code words have an even (or odd) number of ones) has a **Hamming distance of 2**. In this case, a correction to the "correct" code word is not possible.

Error Correction (2)

Obviously:

d = t+1 ⇒ all **<= t bit errors detectable**
(t+1 errors may result in a valid code word.)

d = 2t+1 ⇒ all **<= t bit errors correctable**
decoding to "closest" code word (more bit errors may result in decoding to a wrong code word)

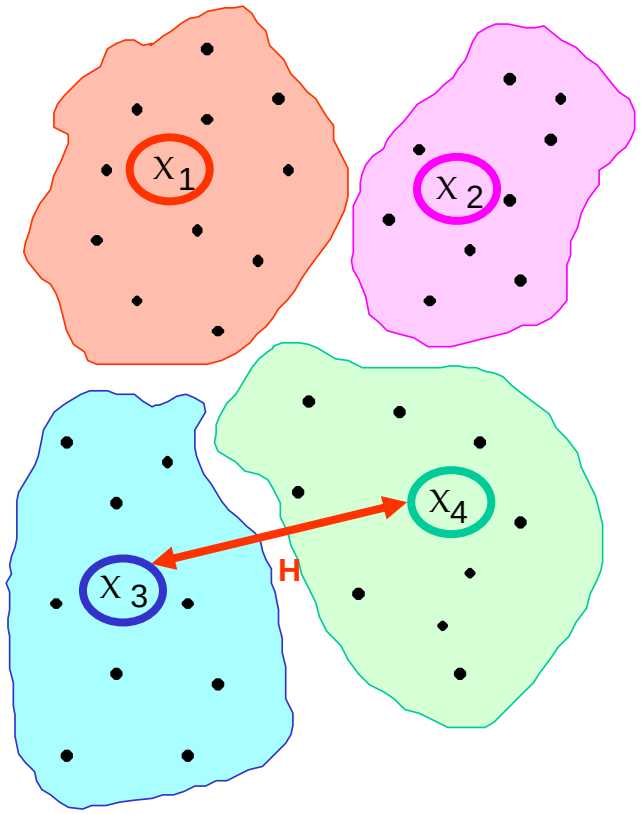
Problems with FEC:

- FEC **reduces** the usable **data rate even when no error occurs**.
- The "closest" code word is **not always correct**.

Therefore:

Only used in special cases

(Simplex channel, real time transmission, …)



X_i = valid bit string

• = bit string with errors, corrected to X_i

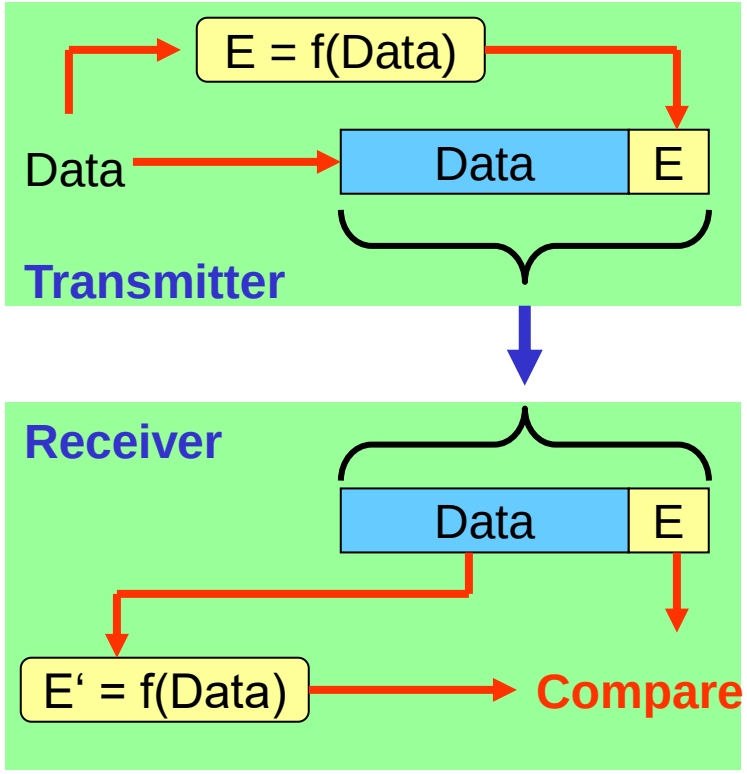
H = Hamming distance

= range of a valid bitstring

Error-Detecting Codes

All error-detection techniques are based on the following principle:

- For a given frame of bits, **additional bits** that constitute an error-detecting code are **added by the transmitter**.
- These additional bits are **calculated as a function of the other transmitted bits**.
- The **receiver performs the same calculation** and compares the two results.
- A detected **error occurs if and only if there is a mismatch**.



The simplest error-detection scheme: Parity bit

- Append a parity bit to the end of a block of data.
- The parity bit is set such that an even number of 1s (even parity) or an odd number of 1s (odd parity) is reached.
- If two (or any even number of) bits are inverted due to error, the error remains undetected.

One of the most common, and one of the most powerful, error-detecting codes is the **cyclic redundancy check (CRC)**.

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Cyclic Redundancy Check, CRC

The cyclic redundancy check is the method **most often used in layer 2** (OSI model).

Idea: The **polynomial** $a_{k-1}\,x^{k-1} + \ldots + a_0$ ($a_i \in GF(2)$) is mapped to a k-bit PDU $(a_{k-1},\ldots,a_0)$.
This means $a_{k-1},\ldots,a_0$ is interpreted as a sequence of coefficients ($\in\{0,1\}$) of a polynomial.

> **Example:** The bit string 110001 represents $x^5 + x^4 + x^0$

Now we use modulo 2 arithmetic, i.e. **binary addition with no carry**. So, addition and subtraction are identical and nothing but the **exclusive-OR operation**.

> **Examples:**
>
> | | | | |
> |---|---|---|---|
> | 10011011 | 00110011 | 11110000 | 01010101 |
> | + 11001010 | + 11001101 | − 10100110 | − 10101111 |
> | 01010001 | 11111110 | 01010110 | 11111010 |

Attention: Division is done by modulo 2 subtraction, too!

(The divisor "fits in" the dividend if the number of bits of the dividend and of the divisor are identical and the highest significant bit of the dividend is "1").

Cyclic Redundancy Check (2)

Usage of the CRC method:

1.) Sender and receiver agree on a **"generator polynomial"**
G(x) = x^r +…+ x^0 with highest and lowest coefficient 1;
represented by the bit string "1...1".

2.) **Map** the PDU to be transmitted (length *m)* to a polynomial ***M(x)***.

3.) The sender extends the PDU by **appending a checksum**. The **resulting polynomial** is **divisible by *G(x)***.

4.) The receiver divides the received message by *G(x).*
Remainder ≠ 0 ⇒ Error detected.

The CRC method is based on a well-defined mathematical framework. This allows us to prove its error-detection capabilities.

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Cyclic Redundancy Check (3)

Chosen as generator polynomial: $G(x) = x^r + \ldots + 1$

corresponds to (r+1) bit string

PDU to be transmitted: $(a_{m-1}, \ldots, a_0)$ — m bit string

Mapping to polynomial *M(x)*: $M(x) = a_{m-1}\, x^{m-1} + \ldots + a_0\, x^0$

Obviously:

$$\frac{M(x) \cdot x^r}{G(x)} = Q(x) + \frac{R(x)}{G(x)}$$

with $R(x) = b_{r-1}\, x^{r-1} + \ldots + b_0\, x^0$

means appending *r* zeros to the PDU

this is an r bit string (one bit less than the "generator string")

quotient

remainder

Cyclic Redundancy Check (4)

We have seen:

$$\frac{M(x) \cdot x^r}{G(x)} = Q(x) + \frac{R(x)}{G(x)} \quad \text{with } R(x) = b_{r-1} x^{r-1} + \ldots + b_0 x^0$$

r bit string

Obviously:

$$\frac{M(x) \cdot x^r - R(x)}{G(x)} = Q(x) = \frac{M(x) \cdot x^r + R(x)}{G(x)}$$

divisible without remainder

due to modulo 2 arithmetic

The checksum to be appended is supposed to be such that the resulting message is divisible by *G(x)* without remainder. This may be achieved as follows:

- **Append *r* zeros** to the PDU to be transmitted,
- **Divide** the resulting string (actually the corresponding polynomial) **by *G(x)*** and
- **Overwrite the** appended ***r* bits by** the calculated **remainder**.

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Cyclic Redundancy Check (5)

Remember: $M(x) = a_{m-1} x^{m-1} + \ldots + a_0 x^0$

$R(x) = b_{r-1} x^{r-1} + \ldots + b_0 x^0$

Transmit: $T(x) = M(x) \cdot x^r + R(x) \longrightarrow (a_{m-1}, \ldots, a_0, b_{r-1}, \ldots, b_0)$

($a_{m-1}, \ldots, a_0$: m bits; $b_{r-1}, \ldots, b_0$: r bits)

Corruption of $T(x)$ may be regarded as addition of an "error polynomial" $E(x)$.

We receive the message $T(x) + E(x)$

Example:

```
    0 1 0 ... 1 1 ... 0      T(x)
⊕   0 0 1 ... 0 1 ... 0      E(x)
-------------------------
    0 1 1 ... 1 0 ... 0      T(x) + E(x)
```

What kind of errors cannot be detected?

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Cyclic Redundancy Check (6)

An error stays undetected if and only if the polynomial corresponding to the received bit string $T(x) + E(x)$ is divisible by the generator polynomial $G(x)$ without remainder:

$$\frac{T(x) + E(x)}{G(x)} = \frac{T(x)}{G(x)} + \frac{E(x)}{G(x)}$$

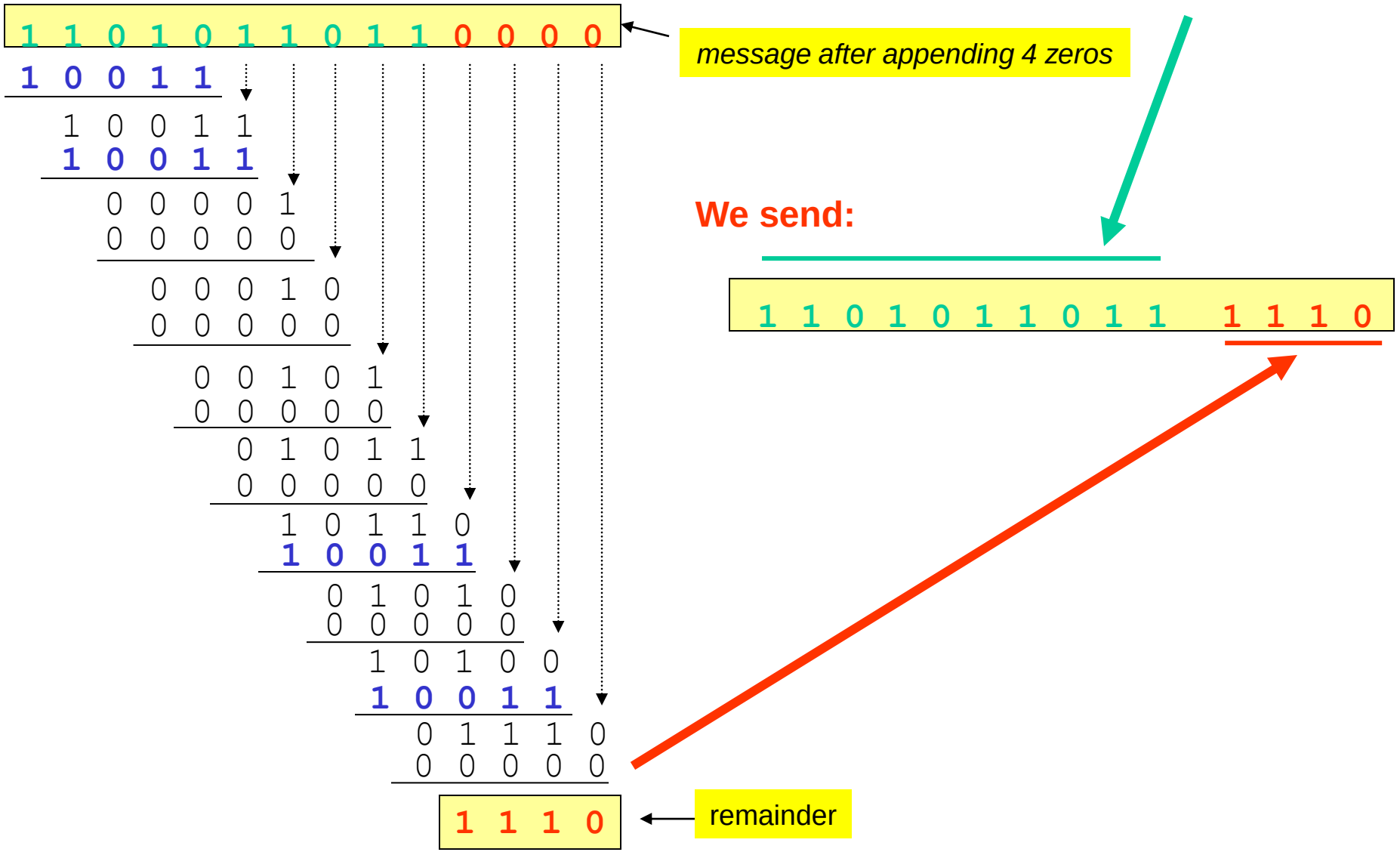
divisible without remainder!

divisible without remainder?

if yes: error stays undetected.

Example: Cyclic Redundancy Check

Generator: 1 0 0 1 1 Message to be sent: 1 1 0 1 0 1 1 0 1 1



Implementation of CRC using Shift Registers

The (non-)existence of the gate follows from the (non-)existence of the term in the generator polynomial.

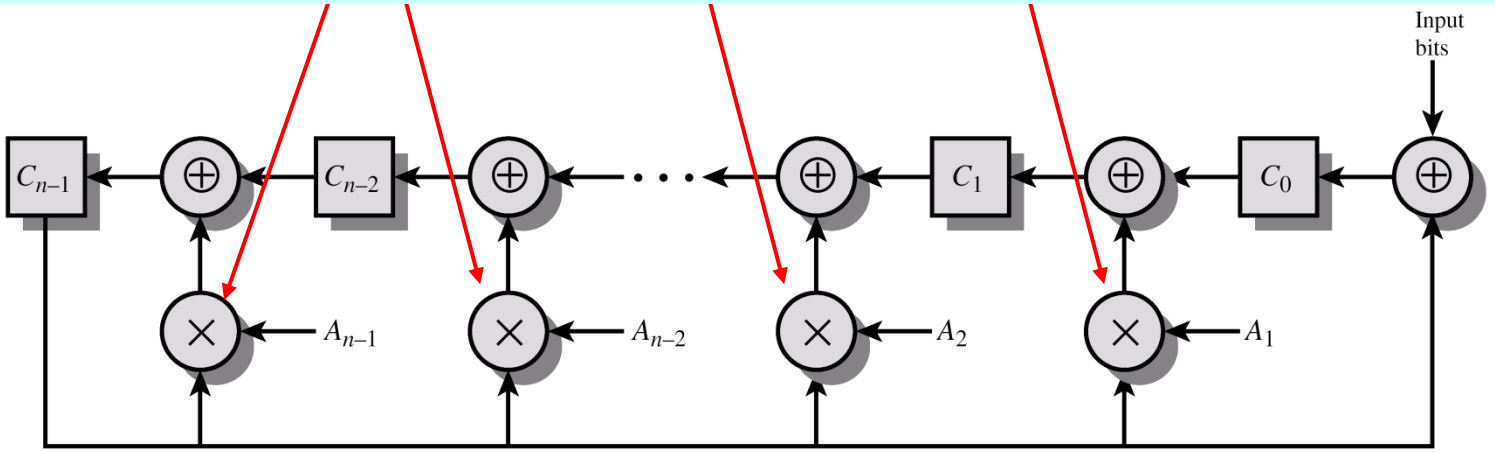
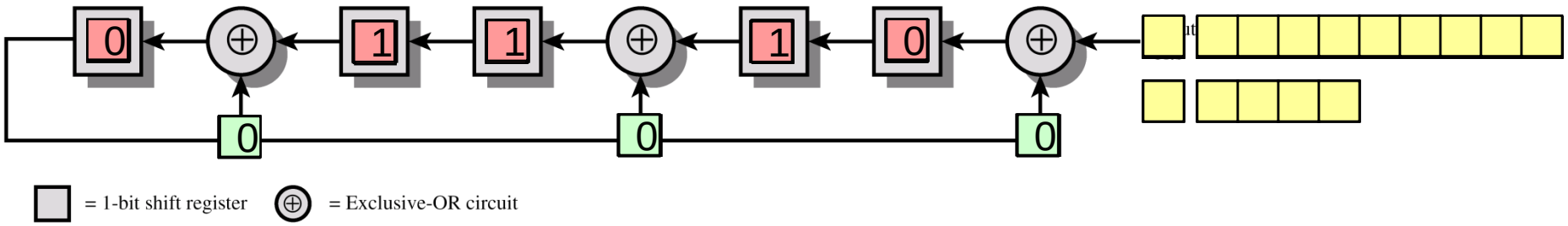


Figure 7.7 General CRC Architecture to Implement Divisor
$1 + A_1X + A_2X^2 + \ldots + A_{n-1}X^{n-1} + X^n$

Source: William Stallings, “Data & Computer Communications”, Prentice Hall, 2000, p. 207

Example: CRC-Implementation with Shift Registers

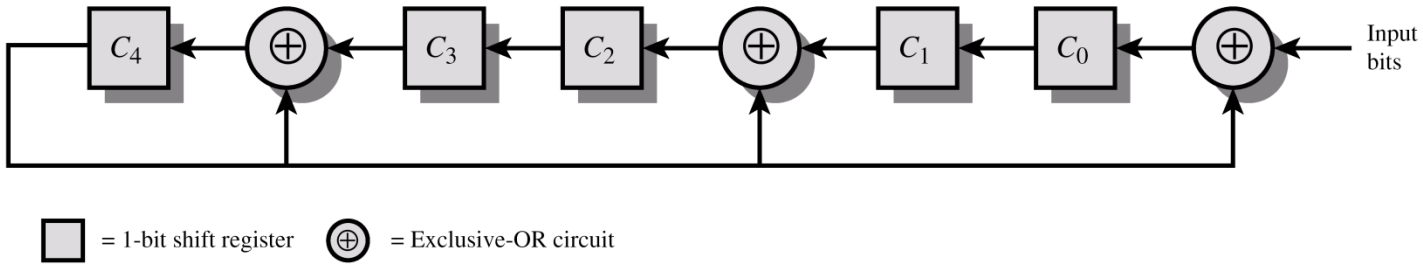
Message: 1010001101 Divisor/Generator: 110101



1. The shifting register contains *n* Bits (*n* = length of the checksum).
2. The XORs fit to the 1s in the generator polynomial.
3. Note: Extend the message with n 0s.
4. After all Bits have run through the register the register contains the remainder.
5. The receiver uses the same implementation.
   The incoming bits are checked „on-the-fly“.
   If there is no error, the remainder is „0...0“.

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Example: CRC-Implementation with Shift Registers (2)



Circuit with Shift Registers for Dividing by the Polynomial $X^5 + X^4 + X^2 + 1$

	C_4	C_3	C_2	C_1	C_0	$C_4 \oplus C_3$	$C_4 \oplus C_1$	$C_4 \oplus$ input	input	
Initial	0	0	0	0	0	0	0	1	1	
Step1	0	0	0	0	1	0	0	0	0	
Step2	0	0	0	1	0	0	1	1	1	
Step3	0	0	1	0	1	0	0	0	0	
Step4	0	1	0	1	0	1	1	0	0	
Step5	1	0	1	0	0	1	1	1	0	Message to be sent
Step6	1	1	1	0	1	0	1	0	1	
Step7	0	1	1	1	0	1	1	1	1	
Step8	1	1	1	0	1	0	1	1	0	
Step9	0	1	1	1	1	1	1	1	1	
Step10	1	1	1	1	1	0	0	1	0	
Step11	0	1	0	1	1	1	1	0	0	
Step12	1	0	1	1	0	1	0	1	0	Five zeros added
Step13	1	1	0	0	1	0	1	1	0	
Step14	0	0	1	1	1	0	1	0	0	
Step15	0	1	1	1	0	1	1	0	—	

(b) Example with input of 1010001101

Source: William Stallings, “Data & Computer Communications”, Prentice Hall, 2000, p. 206

Important Properties of the CRC Method

When choosing appropriate generator polynomials, it may be shown that the following errors (amongst others) will be detected:

a) **All 1 bit errors**

b) **Two isolated 1 bit errors** with (practically) arbitrary large distance.

Example: With $G(x) = x^{15} + x^{14} + 1$ two 1 bit errors with distance < 32768 will be detected.

c) **Odd number** of corrupted bits.

d) **Burst errors** (= ranges with bit errors, possibly with correct bits), if the **burst is shorter than the generator polynomial** (outside the burst: all bits correct).

Common generator polynomials:

CRC-12 $= x^{12} + x^{11} + x^3 + x^2 + x^1 + 1$

CRC-16 $= x^{16} + x^{15} + x^2 + 1$

CRC-CCITT $= x^{16} + x^{12} + x^5 + 1$

If calculating the checksum with CRC-16 or CRC-CCITT, then the following errors will be detected

- all single and double errors,
- all odd bit errors,
- all burst errors of length <= 16
- 99.997 % of 17-bit error bursts
- 99.998 % of bursts with length >= 18 bit.

"Automatic Repeat Request" (ARQ)

When using appropriate codes, only a small percentage of errors remains undetected.

What to do with a corrupted message?

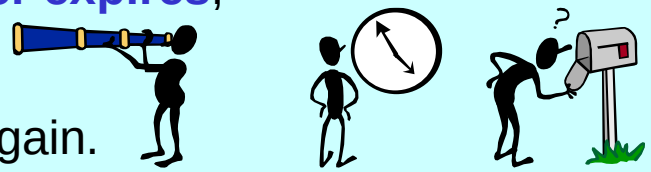
- In general: **Drop** the corrupted message.
- After that: **Retransmission**.

The most important strategy for the detection (and correction) of the loss of entire messages is called **ARQ** (**A**utomatic **R**epeat re**Q**uest).

Idea: After sending a message, the sender waits for a positive acknowledgement.

If the acknowledgement does not arrive before **a timer expires**,

- the message is **repeated** and
- the sender **waits for** a positive acknowledgement again.

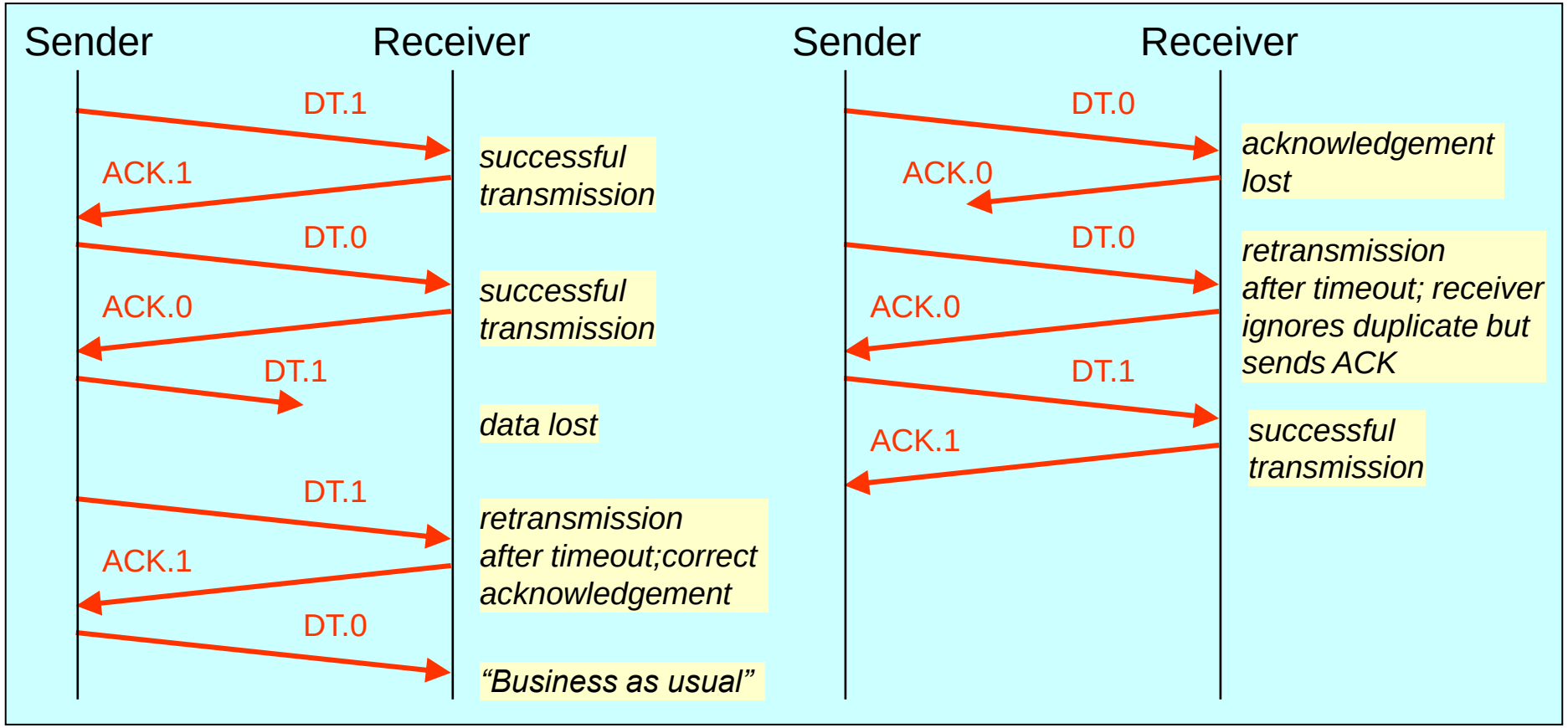


A “new” message is only transmitted after the acknowledgement for the previous message has been received by the sender.

This strategy is called **"stop-and-wait"**.

The Alternating Bit Protocol

The “alternating bit protocol” is a straight-forward way of realizing stop-and-wait:



Obviously, message numbering is based on **1 bit only**.

In real life, this approach is used in short-range networks such as Bluetooth.

When sending data in both directions, the **acknowledgement may be carried together with the data in the opposite direction** ("piggybacking").

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Repetition of Acknowledgements

What to do when losing messages or acknowledgements?

The sender repeats the message

a) until an acknowledgement is received or
b) until communication is terminated

The receiver may repeat acknowledgements, if message flow stops unexpectedly.

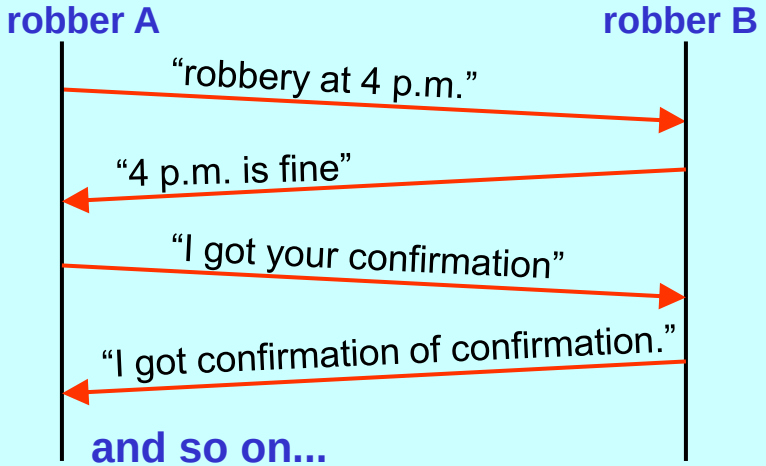
But:
Completely reliable communication via unreliable channels is not possible!

Example:

Two robbers communicate via a messenger (who may get lost/caught).

Goal: Arrange the time of a robbery

Problem: The last message is not protected.
All the other messages get confirmed.



Relevance to computer networks: Problems with teardown of connections.

Important Parameters

The performance of the stop-and-wait protocol depends on several protocol parameters and on the “environment”.

- A: Acknowledgement length (in bit)
- C: Channel capacity (in bit/s)
- D: Data field length (in bit)
- E: Probability of bit errors (attention: independency of errors assumed)
- F: D + H (length of PDU, in bit)
- H: Header length (in bit)
- I: Interrupt service time and signal delay (in s)
- L: Probability of losing a data PDU or its acknowledgement
- P_1: Probability of losing a data PDU
- P_2: Probability of losing an acknowledgement
- R: Average number of repetitions of a data PDU
- T: Timeout interval (in s)
- U: Utilization of the channel (efficiency)

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Channel Utilization of Stop-and-Wait with Error-free Transmission

In case of transmission without errors: — sending first to last bit

F/C = time for **transmission** of a **data PDU** (length divided by data rate)

(F/C) + I = time until the **transmission** of the **acknowledgement starts**.

- F/C: sending time
- I: service time + signal delay

At time (F/C) + I + (A/C) + I sender is able to send **next data PDU**.

- (F/C) + I: from sender
- (A/C) + I: back to sender

The **efficiency** may be calculated as:

$$\frac{\text{used time}}{\text{total time}} = \frac{\text{\# data bits sent}}{\text{total time} \bullet C} = \frac{D}{H+D+A+2CI} = \underbrace{\frac{D}{H+D}}_{\text{header overhead}} \bullet \underbrace{\frac{1}{1+\frac{A+2CI}{H+D}}}_{\text{waiting due to stop-and-wait}}$$

Channel Utilization with Error-free Transmission (continued)

What is the maximum percentage of the raw bit rate really available to the application ?

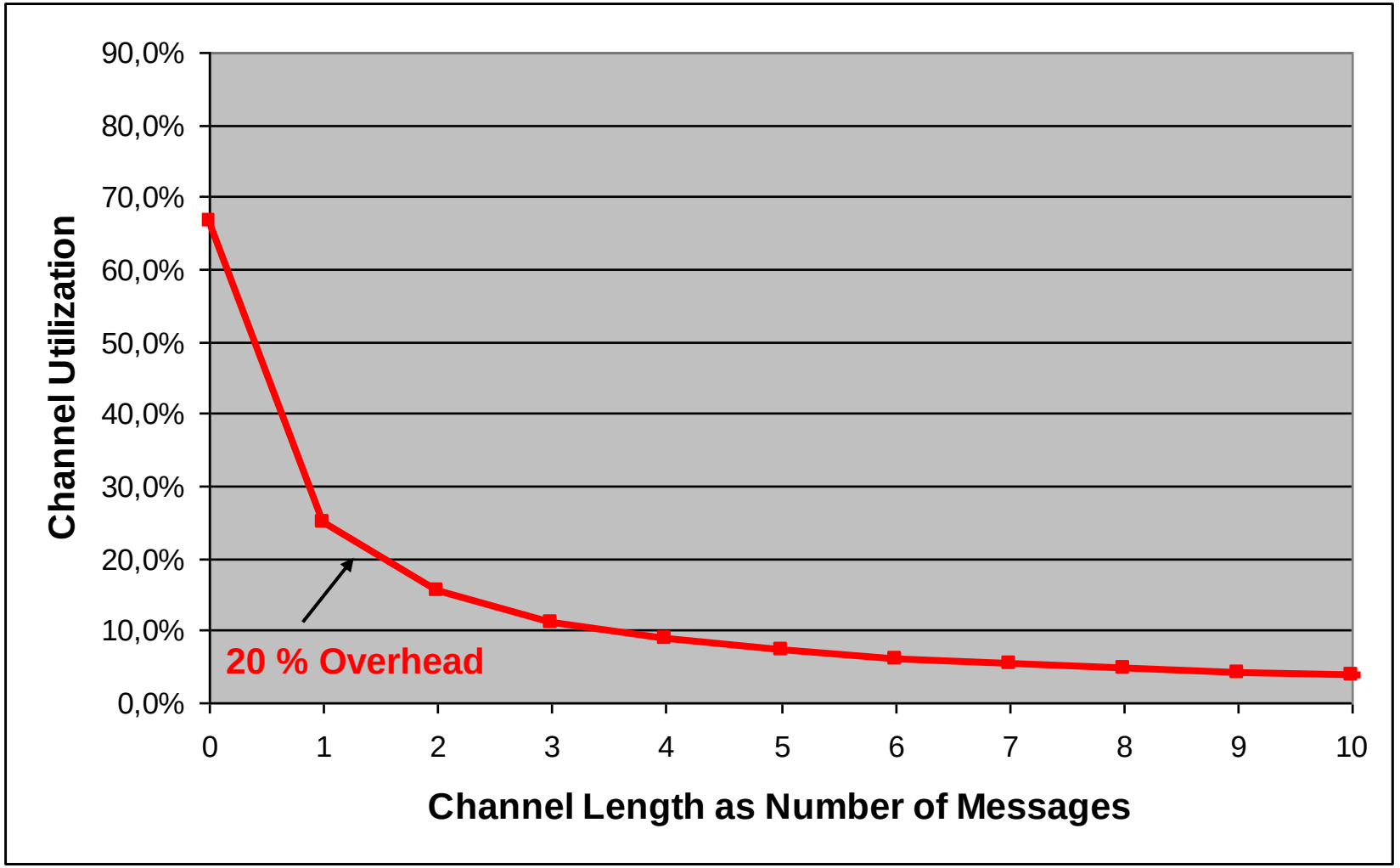
Choose the parameters and calculate !

Is this data rate really available ?

$$\frac{\text{used time}}{\text{total time}} = \frac{\text{\# data bits sent}}{\text{total time} \bullet C} = \frac{D}{H + D + A + 2CI}$$

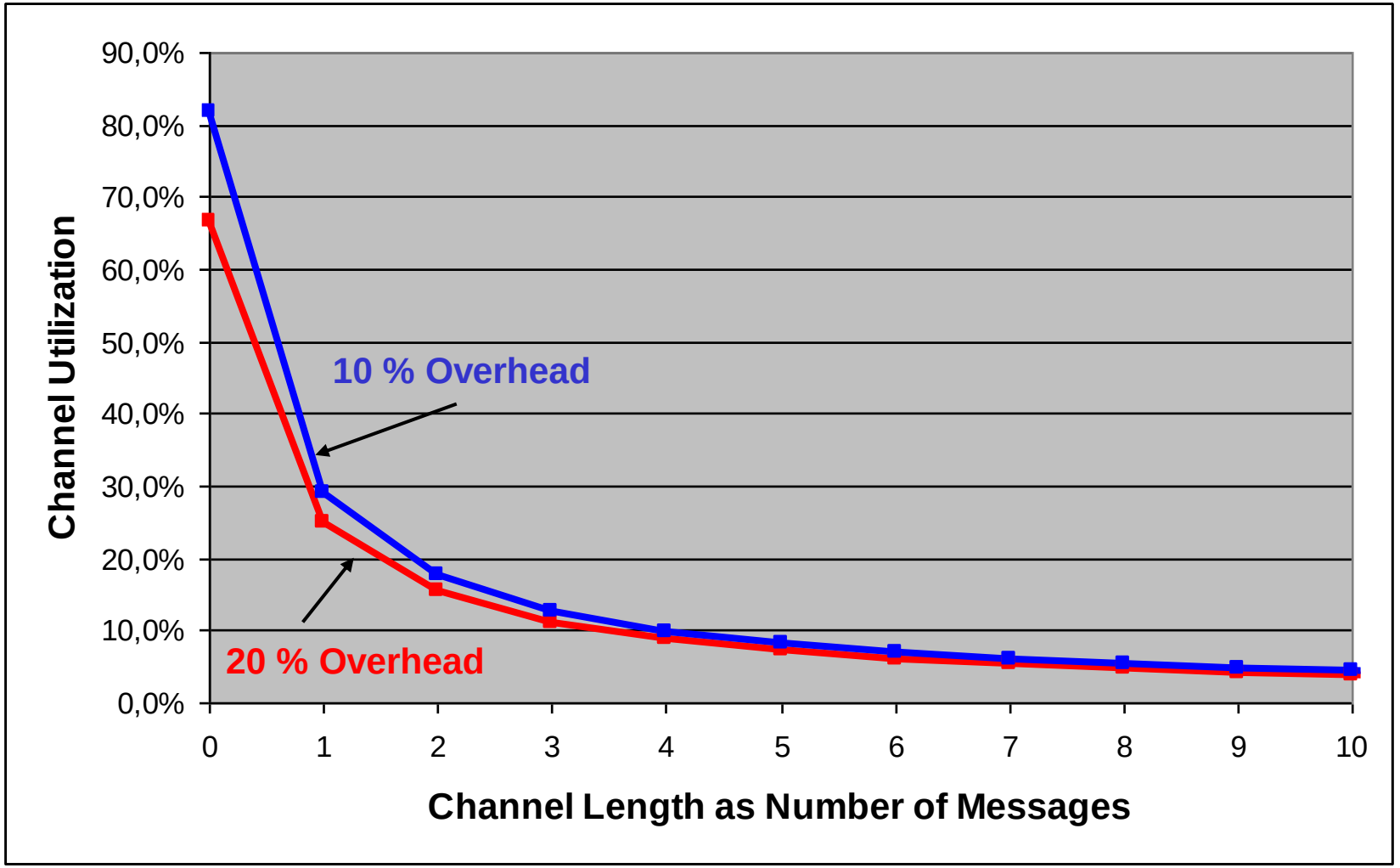
Channel Utilization with Error-free Transmission (continued)

With 10 % and 20 % message overhead (D/(H+D) = 90 % and 80 %, respectively) we find the following results:



Channel Utilization with Error-free Transmission (continued)

With 10 % and 20 % message overhead (D/(H+D) = 90 % and 80 %, respectively) we find the following results:



Channel Utilization with Transmission Errors

If the sender detects an error with **timeout only**, an **unsuccessful transmission** consumes the transmission time of F + CT bits. $= C \cdot (F/C + T)$

sending *waiting*

The average total time for a successful transmission may be calculated as

R · failure *1 · success*

$$\frac{R(F+CT)+(F+A+2Cl)}{C}$$

Question: **How large is R**, the average number of duplicates per successful transmission?

The probability of an unsuccessful transmission is

probability: both successful

$$L = 1 - (1 - P_1)\,(1 - P_2)$$

success data *success ACK*

Assuming geometric distribution, (indep. attempts, same error prob.), this yields : $R = L/(1 - L).$

The channel utilization is:

$$U = \frac{D}{\frac{L}{1-L}(F+CT)+(F+A+2Cl)}$$

$= R$

Reminder: Geometric Distribution

Observe so-called **"Bernoulli trials"**, that is independent experiments with

p : **success** probability
q = 1-p : **failure** probability

Let **X** be the **number of experiments until the first success**. The random variable X **has geometric distribution**:

$$P\{X=n\} = \underbrace{(1-p)^{n-1}}_{\text{probability for (n-1) failures}} \underbrace{p}_{\text{probability for success in n-th trial}} \; ; \quad n=1,2,\ldots$$

Expected/avg. value:

$$E[X] = \sum_{n=1}^{\infty} n\, p\, (1-p)^{n-1} = \ldots = \frac{1}{p}$$

Substitute **"p"** by **"1-L"** to study the channel utilization. This yields:

Average number of (sender) transmission repetitions:

$$R = E[X] - 1 = \frac{1}{1-L} - \frac{1-L}{1-L} = \frac{L}{1-L}$$

success

Channel Utilization with Transmission Errors (continued)

For **T = A/C + 2I**, that is constant interrupt service time and **minimal timeout value**, we get:

sending all bits of acknowledge-ment

signal delays and processing time of sender and receiver

$$U = \underbrace{\frac{D}{H+D}}_{\text{header overhead}} \bullet \underbrace{(1-P_1)(1-P_2)}_{\text{unsuccessful transmission}} \bullet \underbrace{\frac{1}{1+\frac{CT}{H+D}}}_{\text{waiting due to stop-and-wait}}$$

Independency and A = H leads to

$$U = \frac{D}{H+D} \bullet \underbrace{(1-E)^{H+D}}_{\text{All bits of PDU successful}} \underbrace{(1-E)^{H}}_{\text{All bits of ACK successful}} \bullet \frac{1}{1+\frac{CT}{H+D}}$$

Question: Which packet size yields optimum efficiency?

Channel Utilization with Transmission Errors (continued)

Question: Which packet size yields optimum efficiency?

Answer: Differentiate with respect to D and set derivation to 0.

Result:

$$D_{opt} = \frac{H + CT}{2}\left(\sqrt{1 - \frac{4}{(H + CT)\ \ln(1 - E)}} - 1\right)$$

For low bit error probability, this is approximately

$$D_{opt} \approx \sqrt{\frac{H + CT}{E}}$$

It is easy to see:

large E $\Rightarrow$ choose **small D** (reduces risk of loss)

small E $\Rightarrow$ choose **large D** (reduces influence of header and stop-and-wait)

Channel Utilization with Transmission Errors (continued)

What is the maximum percentage of the raw bit rate really available to the application ?

Choose the parameters and calculate !

Is this data rate really available ?

$$U = \frac{D}{H+D} \bullet \underbrace{(1-E)^{H+D}}_{\text{All bits of PDU successful}} \underbrace{(1-E)^{H}}_{\text{All bits of ACK successful}} \bullet \frac{1}{1+\frac{CT}{H+D}}$$

3.5. Multiple Access Schemes

Goal: parallel wireless communication of several devices in spatial proximity.

Required: controlled separation of the communication

Time Division Multiple Access (TDMA):
- all communication on the same frequency
- separation by allocating time slots for communication

Frequency Division Multiple Access (FDMA):
- all communication at the same time
- separation by allocating different frequencies for communication

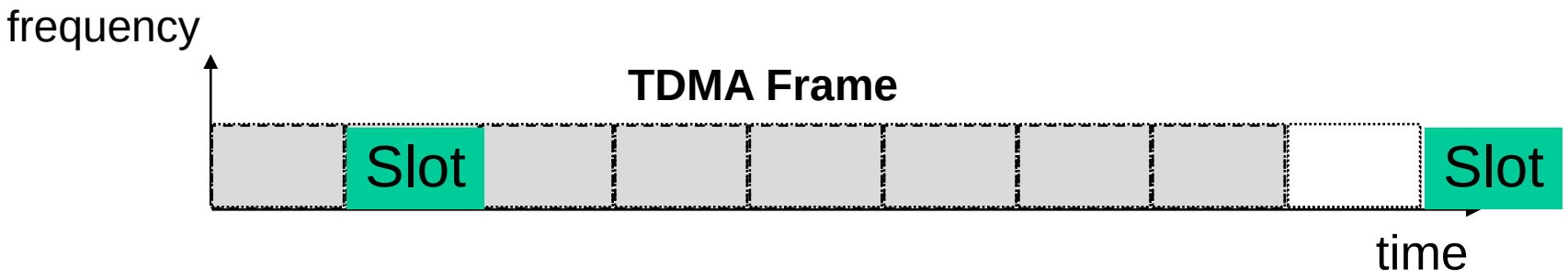
Code Division Multiple Access (CDMA):
- all communication at the same time on the same frequency
- separation by allocating different communication codes

Space Division Multiple Access (SDMA):
- all communication at the same time on the same frequency
- separation by allowing separating the communication based on the users' location

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Time Division Multiple Access

Time Division Multiple Access (TDMA):
- each channel is split up into time slots
- time slots repeat periodically (also called TDMA frames)
- each sender is assigned a fixed subset of slots per frame



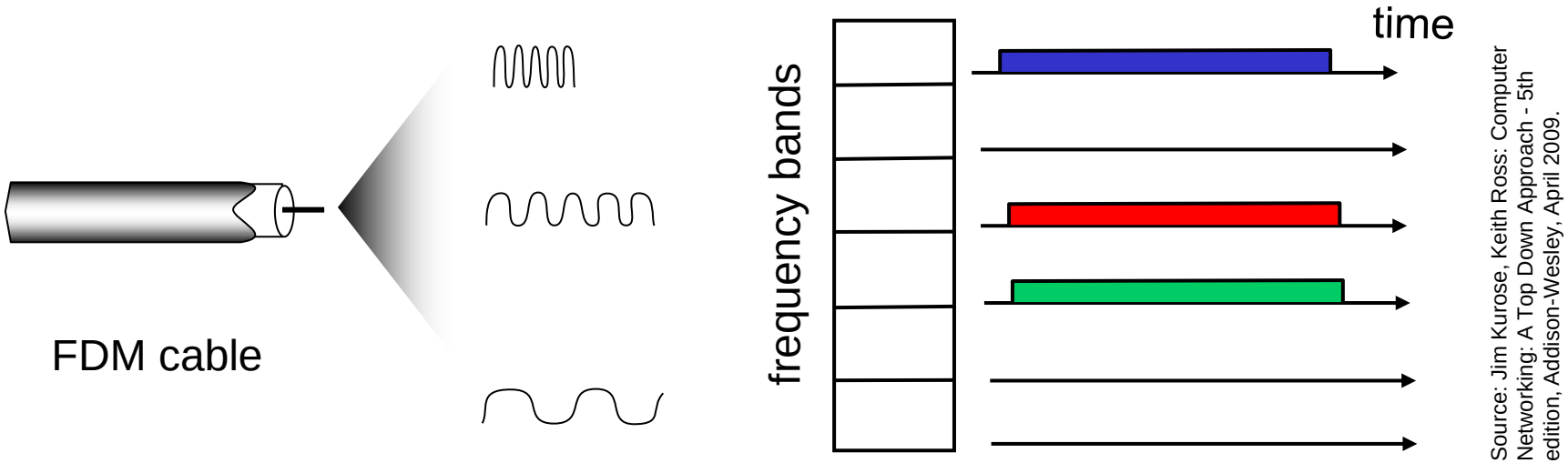
Properties:
- different users transmit/receive in different time slots
- requires exact synchronization of the devices
- devices may be turned off during idle slots to save power

Challenge:
- mobile devices are placed at different distances to each other (different round trip times)
- mobile devices change their location (round trip times change)
- a scheduling policy is required

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Frequency Division Multiple Access

Frequency Division Multiple Access (FDMA):
- each channel is split up into narrower frequency bands, so-called **sub-carriers**
- each sub-carrier is used exclusively by one user



Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

Note: Perfect separation of sub-carriers is impossible due to imperfect filters!
- * **Guard-Bands** between neighbouring frequencies
  - this limits the max. achievable data rate of a channel

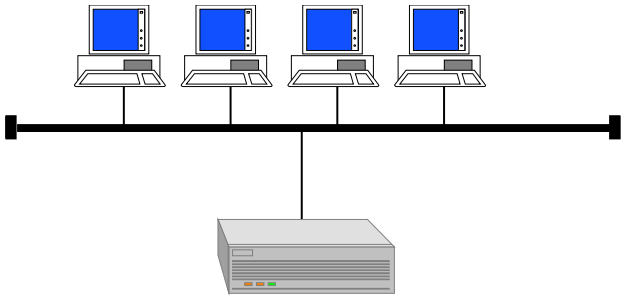
Typical usage: FDD (Frequency Division Duplex)
Base station and mobile use two different frequencies to achieve a full duplex communication.

3.6. Medium Access Control

When connecting **several stations to a shared medium**, **medium access control** becomes essential.

IEEE 802 developed several important protocols for local area networks (LANs).

In LANs, we may use specific mechanisms **not suitable for larger distances**.



3.6.1. Aloha from Hawai

3.6.2. The classical "Local Area Network"

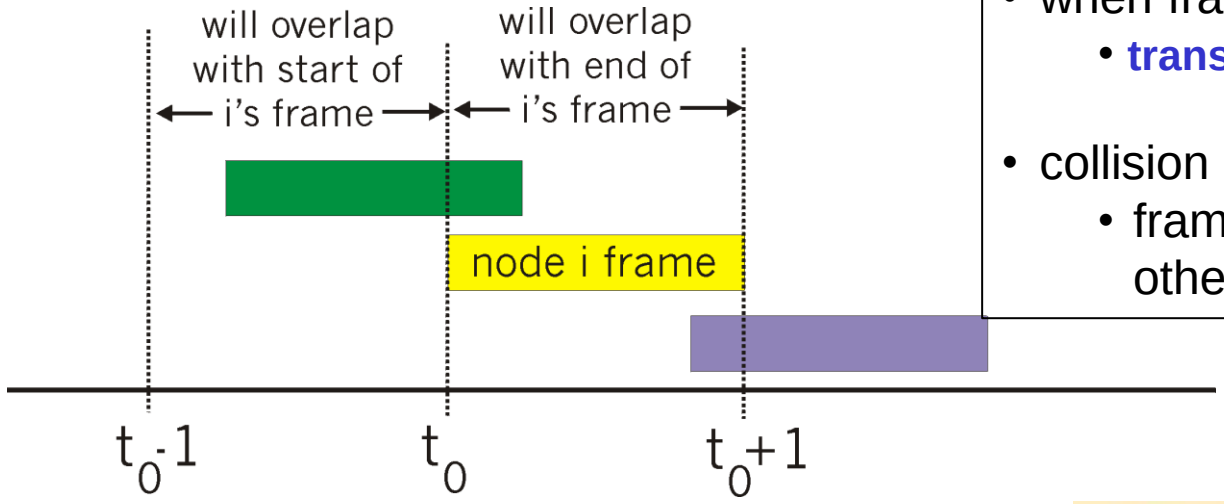
3.6.3. Ethernet

3.6.4. WLAN … in a Nutshell

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3.6.1. Aloha from Hawaii

Aloha: Start a transmission any time you like.



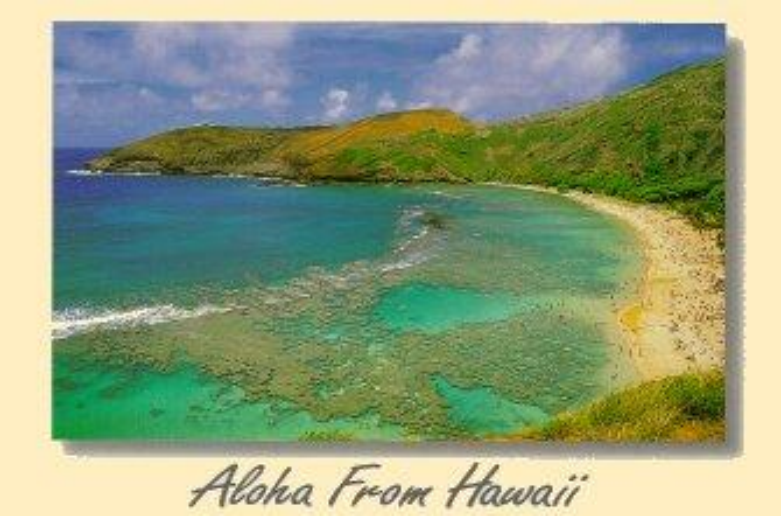
- when frame first arrives
  - **transmit immediately**
- collision probability increases:
  - frame sent at t_0 collides with other frames sent in $[t_0-1,t_0+1]$

Advantages of Pure Aloha:

- **Simple mechanism, easy to implement**
- **Illustration of general random access network characteristics**
- **Easy to analyze**

Disadvantage of Pure Aloha:

- **Low maximum throughput**



Assumptions …

Assumptions for our analysis:

- **Error-free channel** (frames lost by collisions only)
- **Retransmission in case of missing ACK** after random backoff time
- **Fixed frame length *P*** (in seconds, “P” like “packet”)
- **Complete arrival process** (incl. retransmissions) **Poisson Arrivals**
  (Model of infinite user population with very small individual arrival rates)

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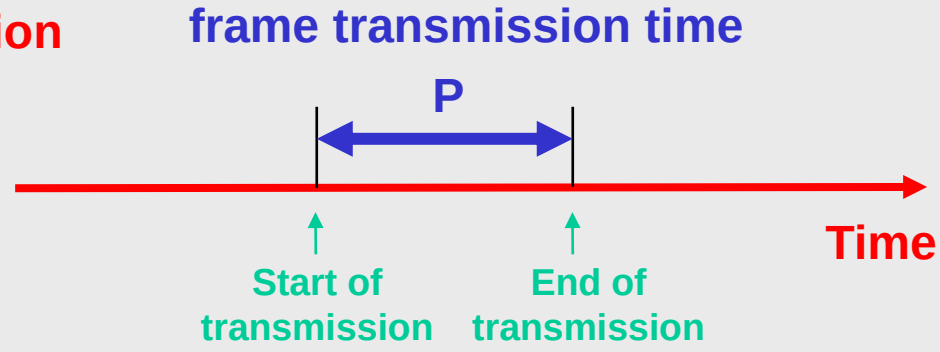
Behavior of a Reference Scenario

We define throughput S and offered load G as follows:

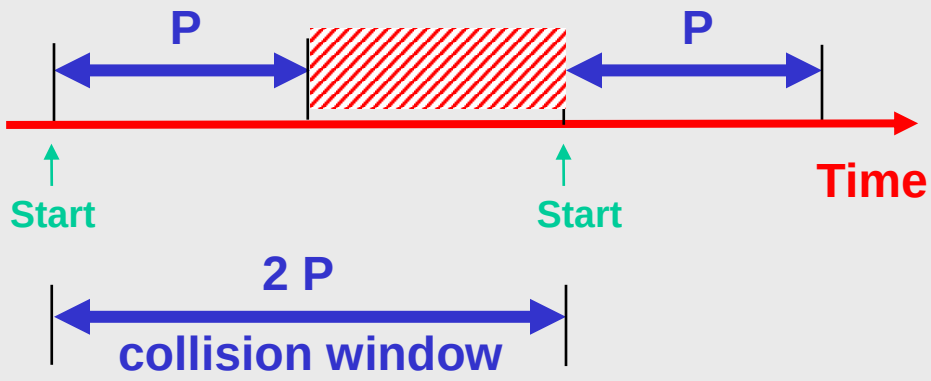
S: Average number of **successful transmissions per frame transmission time *P***
G: Average number of **attempted transmissions per frame transmission time *P***

Now consider the behavior of a **reference station**:

Transmission by the reference station



Transmission by other station



All transmission attempts starting during the collision window result in a collision

Throughput versus Offered Load for Aloha

Obviously, S may be calculated as

S = G · P {successful transmission}
S = G · P {0 arrivals in interval of length 2P}

In case of **Poisson Arrivals** we find:

S = G · e^{-2G}

$$P\{\text{k arrivals in T seconds}\} = \frac{(\lambda T)^k e^{-\lambda T}}{k!}, k = 0, 1, 2, \ldots$$

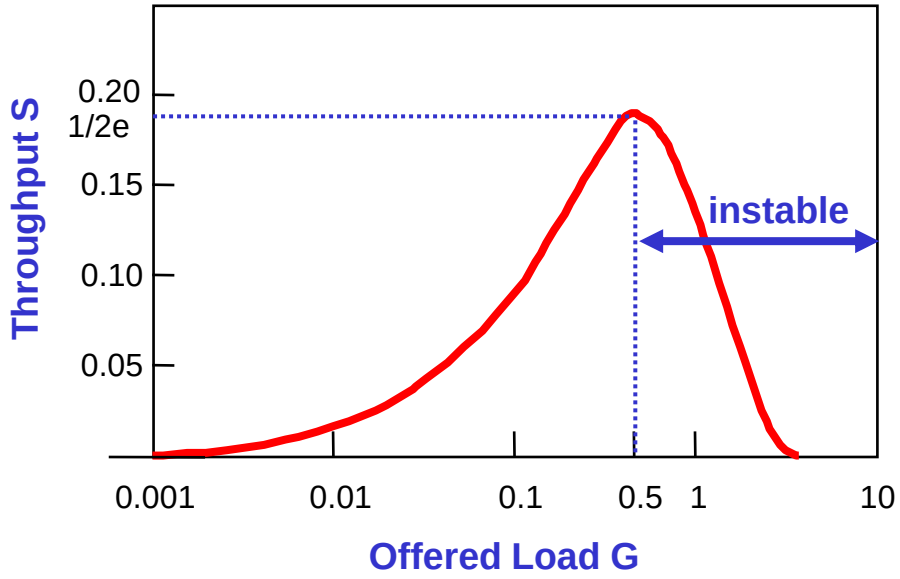
$$P\{\text{0 arrivals in T seconds}\} = \frac{(\lambda T)^0 e^{-\lambda T}}{0!} = e^{-\lambda T}$$

λ T = average number of arrivals in interval of length T

2 G = average number of arrivals in interval of length 2P

It may easily be shown that the **maximum throughput is reached for G = 0.5**.

In this case: **S = 1/2e ≈ 0.184** < 20 %



Note:
In the **instable area**
- **additional load** results in
- **throughput degradation**

Slotted Aloha (originally developed for satellite communication)

Obviously:

- Pure Aloha is **instable for G > 0.5.**
- The **throughput** for Pure Aloha is **< 20 %.**

Slotted Aloha **improves the performance by** segmenting the time into

- **slots of fixed length P**
- exactly **equal to the frame transmission time**

Idea: **Delay frame transmissions** until the start of the next slot

Result: **Frames either overlap completely or not at all**,
i.e. the collision window is reduced by 50 %.

Now the throughput S may be calculated as

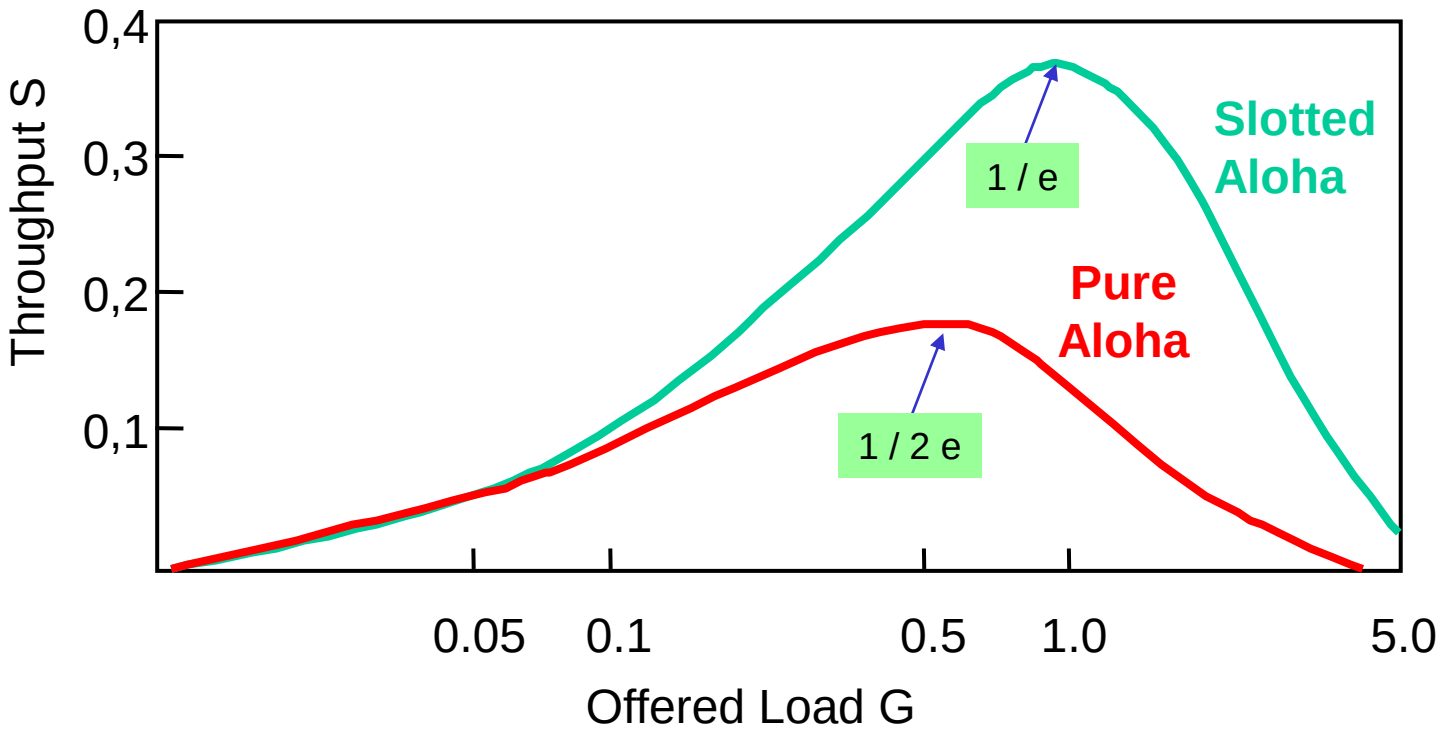
S = G · P {successful transmission}
S = G · P {0 arrivals in interval of length P}

instead of 2P

Performance Comparison: Pure Aloha - Slotted Aloha

Because of Poisson arrivals: $S = G \cdot e^{-G}$ (instead of $S = G \bullet e^{-2G}$)

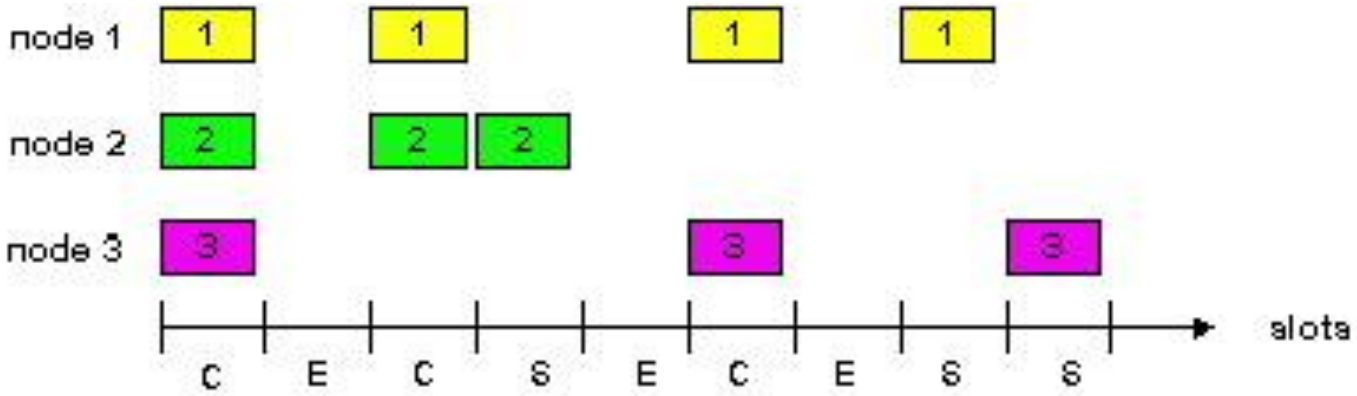
The maximum throughput is $S = 1/e \approx 0.368$ (for $G = 1$).



Slotted Aloha achieves **higher throughput** and **(slightly) improved stability**.

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Slotted Aloha



Assumptions:
- all frames same size
- time divided into equal size slots (time to transmit 1 frame)
- nodes start to transmit only at slot beginning
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

Operation:
- when node obtains fresh frame, transmits in next slot
  - if no collision: node can send new frame in next slot
  - if collision: node retransmits frame in each subsequent slot with prob. p until success

3.6.2. The classical "Local Area Network"

Until the middle of the 1990s, the shared medium was thought to be characteristic for LANs:

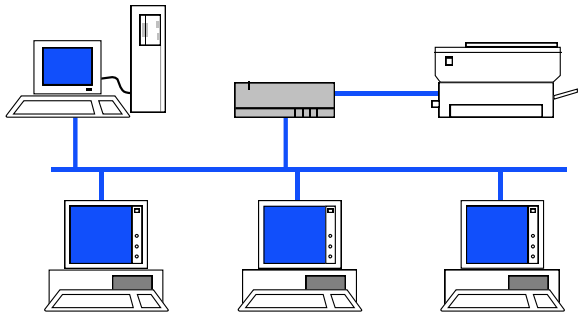
Key Concepts

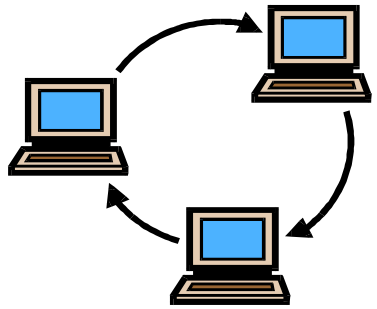
The LANs described herein are distinguished from other types of data networks in that they are **optimized for a moderate-sized geographic area**, such as a single office building, a warehouse, or a campus.

The IEEE 802 LAN is a **shared-medium peer-to-peer communication network that broadcasts information** for all stations to receive...

There is always need for an **access sublayer** in order to arbitrate the access to the shared medium.

From: IEEE Std. 802-1990; IEEE Standards for Local and Metropolitan Area Networks: Overview and Architecture





IEEE 802 Home Page

ieee802.org

IEEE 802 LMSC

https://www.ieee802.org/

The IEEE 802 LAN/MAN Standards Committee (LMSC) develops and maintains networking standards and recommended practices for local, metropolitan, and other area networks, using an open and accredited process, and advocates them on a global basis. The most widely used standards are for Ethernet, Bridging and Virtual Bridged LANs Wireless LAN, Wireless PAN, Wireless MAN, Wireless Coexistence, Media Independent Handover Services, and Wireless RAN. An individual Working Group provides the focus for each area.

Get 802 standards

09.11.2026

- Get 802 Standards
- (IEEE-SA / IEEE 802 LMSC 2016 "Get IEEE802" Program Guideline)

IEEE 802 LMSC Working Groups, Technical Advisory Groups, Study Groups, and Standing Committees

- 802.1 Higher Layer LAN Protocols Working Group
- 802.3 Ethernet Working Group
- 802.11 Wireless LAN Working Group
- 802.15 Wireless Specialty Network (WSN) Working Group
- 802.18 Radio Regulatory TAG
- 802.19 Wireless Coexistence Working Group
- 802.24 Vertical Applications TAG
- IEEE 802 LMSC Standing Committees
- List of Hibernating and Disbanded Working Groups and Study Groups

IEEE 802 LMSC Session Information

- IEEE 802 LMSC Plenary/Interim meeting information
- IEEE 802 LMSC Meeting Agendas, Minutes and Reports
- PARs under consideration
- Tutorial information for the plenary meetings
- IEEE 802 LMSC conference call schedule and minutes
- IEEE 802 LMSC and subgroup calendar

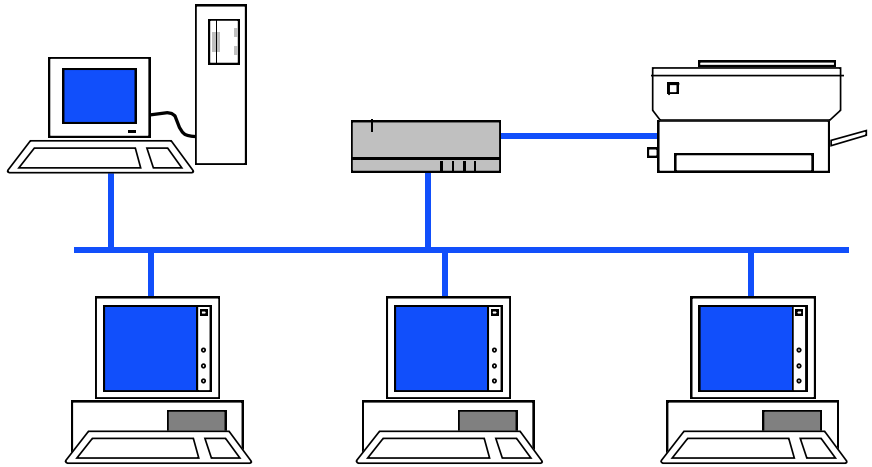
IEEE 802 LMSC General Information

- IEEE SA Policies
- IEEE SA Training
- Approved/In process IEEE 802 Activities (Standards/Projects/Pre-par)
- IEEE 802 LMSC Orientation
- IEEE 802 LMSC Policies & Procedures and other documents related to standards development
- List of active IEEE 802 Standards
- IEEE 802 LMSC Archived and Obsolete Documents
- Join a ballot pool for 802 standards (and other IEEE standards)
- Appeals made to the 802 Executive Committee
- IEEE 802 LMSC Liaisons
- IEEE 802 LMSC Public Statements to Government Bodies
- IEEE Computer Society Home Page
- Standards Development Toolbox
- 802 Wireless Frequency Band Table
- Participation in IEEE activities by Participants on the BIS Entity List
  - FAQs
  - IEEE Statement
  - IEEE SA Statement
- Draft Sharing Status

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3.6.3. Ethernet

1973: At XEROX-PARC, people are thinking about networks for early PCs.



Bob Metcalfe's idea:

- All devices use the same cable.
- Listen and check whether line is free.
  If yes: Try to transmit.
- Collisions are detected and
  solved by randomized waiting.

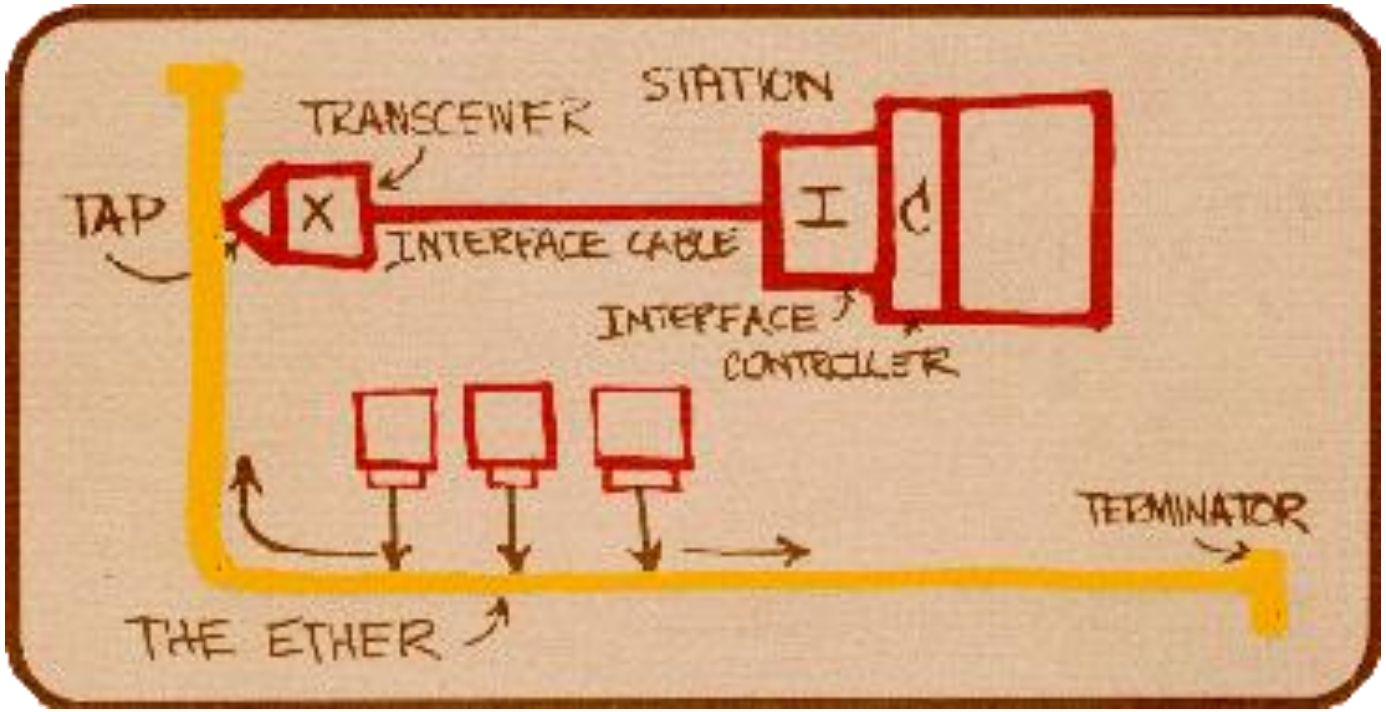
Since 1980: Ethernet is commercially available.

C	**Carrier**
S	**Sense**
M	**Multiple**
A	**Access with**
C	**Collision**
D	**Detection**

Ethernet: Communication must be inexpensive

1976:

Robert M. Metcalfe draws a diagram for the presentation of “Ethernet” at the National Computer Conference in June 1976.



“Like the computing stations to be connected, the communication system must be inexpensive.”

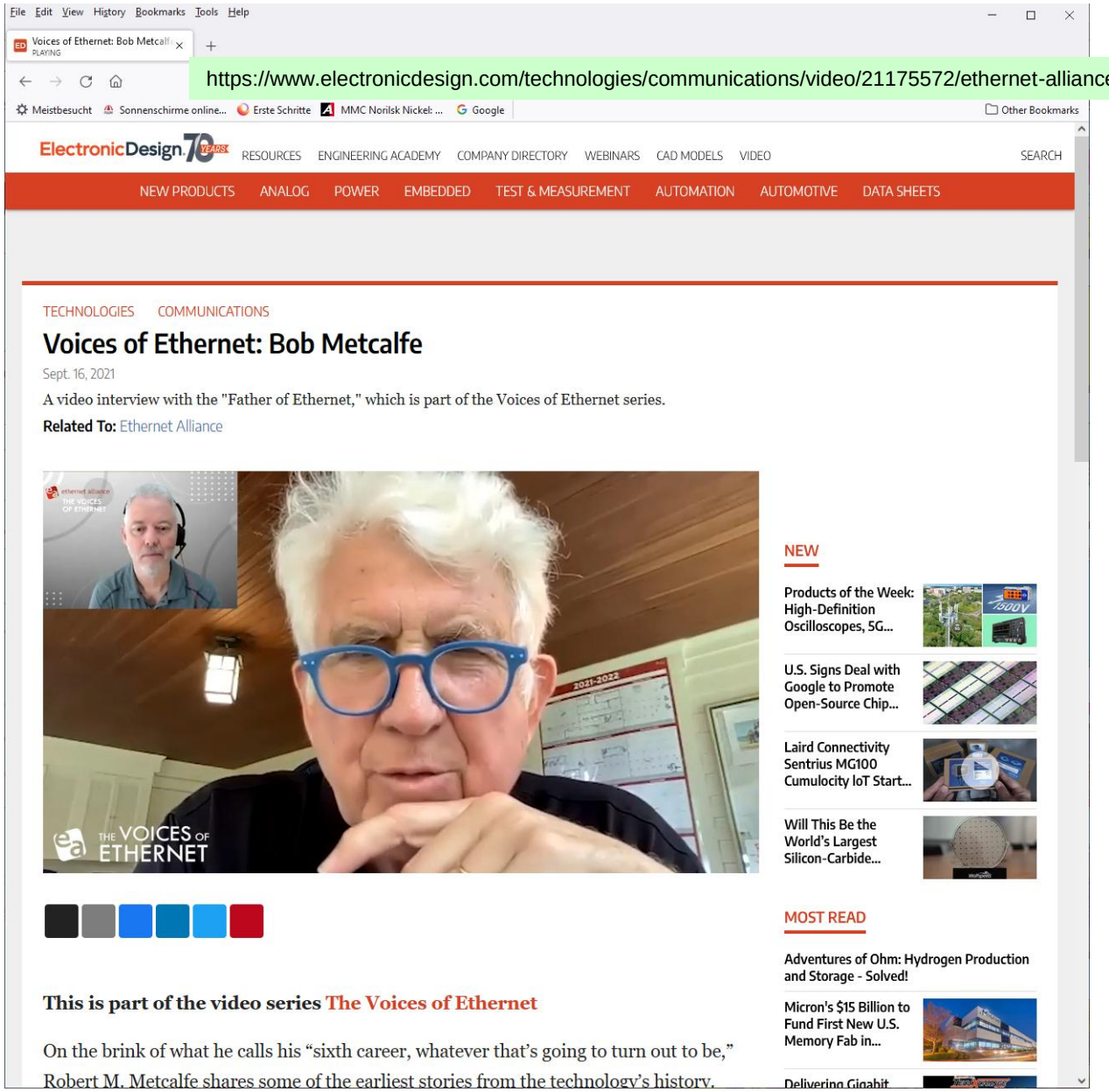
R. M. Metcalfe, D.R. Boggs, “Ethernet: Distributed Packet Switching for Local Computer Networks”, Communications of the ACM, Vol. 19, No. 5., July 1976, pp. 395 - 404

“The Ether is a logically passive medium for the propagation of digital signals and can be constructed using any number of media including coaxial cables, twisted pairs, and optical fibers.”

R. M. Metcalfe, D.R. Boggs, “Ethernet: Distributed Packet Switching for Local Computer Networks”, Communications of the ACM, Vol. 19, No. 5., July 1976, pp. 395 - 404

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Recommendation: Listen to Bob Metcalfe; Interview on Sept. 16, 2021



"CSMA/CD"

This strategy of Ethernet and IEEE 802.3 LANs allows **immediate medium access** if no other station uses the **shared bus**.

Collisions are detected and resolved:

C	**Carrier**
S	**Sense**
M	**Multiple**
A	**Access with**
C	**Collision**
D	**Detection**



Same as:
"Listen-Before-Talk" in conversations.

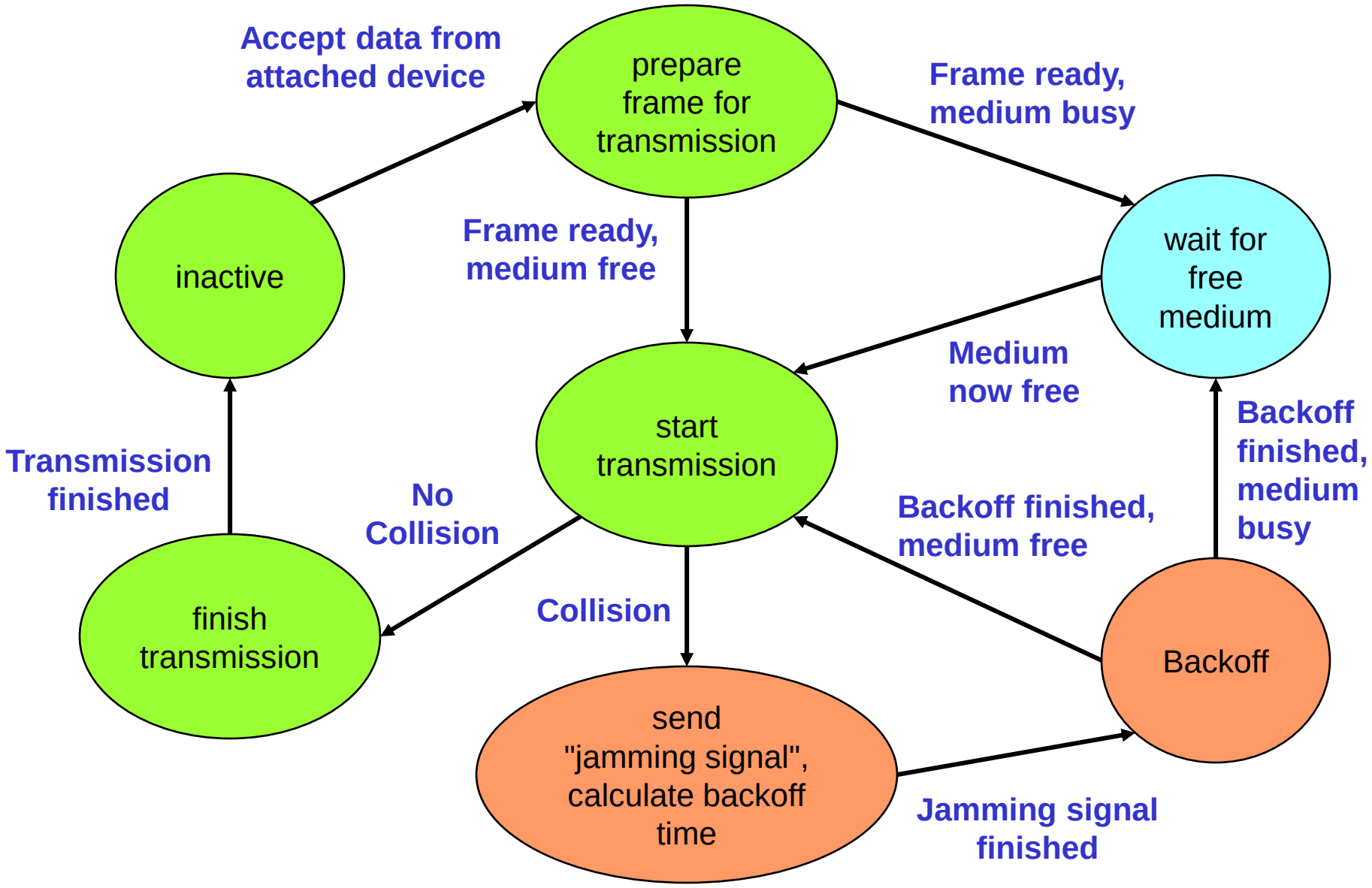
CSMA/CD is based on a protocol called **"Aloha"** which was developed in Hawaii for satellite communication.

Aloha: - Send whenever you want
- Collisions are detected by message loss (missing ACK)
- Lost data are re-transmitted

In LANs, the signal propagation time is small. For this reason, **collisions may be detected while transmitting**.

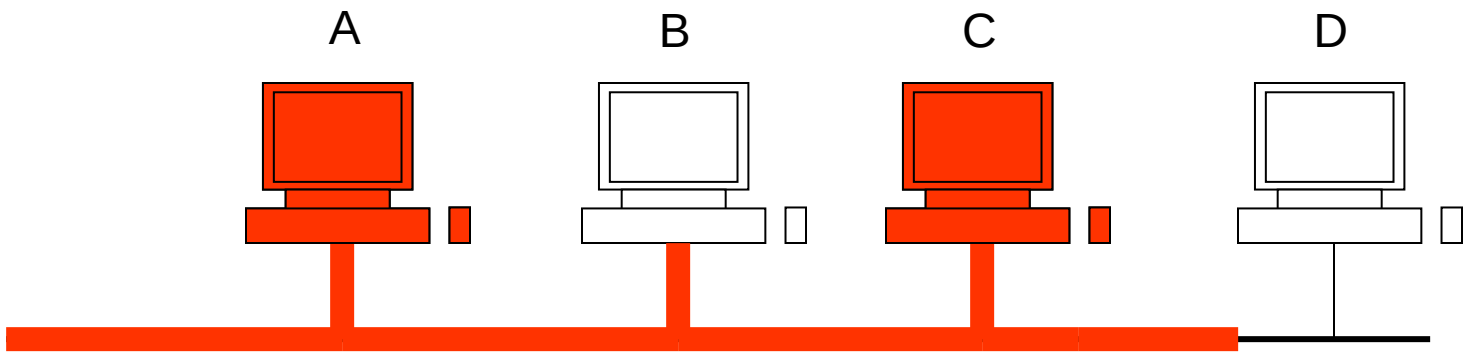
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CSMA/CD medium access



Carrier Sense Multiple Access with Collision Detection

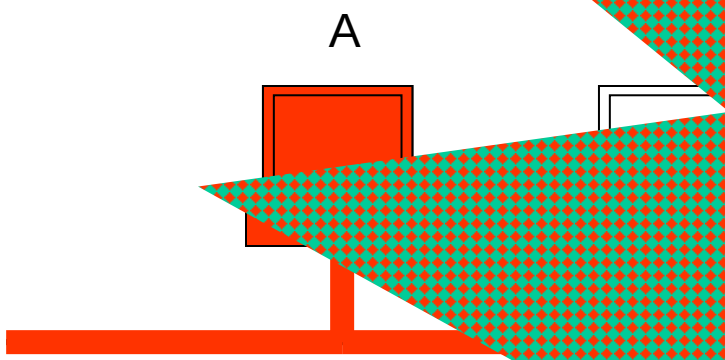
Classical Ethernet controls medium access with "CSMA/CD":



1. Station A wants to transmit to station C and senses a free medium.
2. Station A starts the transmission.
3. Station C receives the beginning of the message.

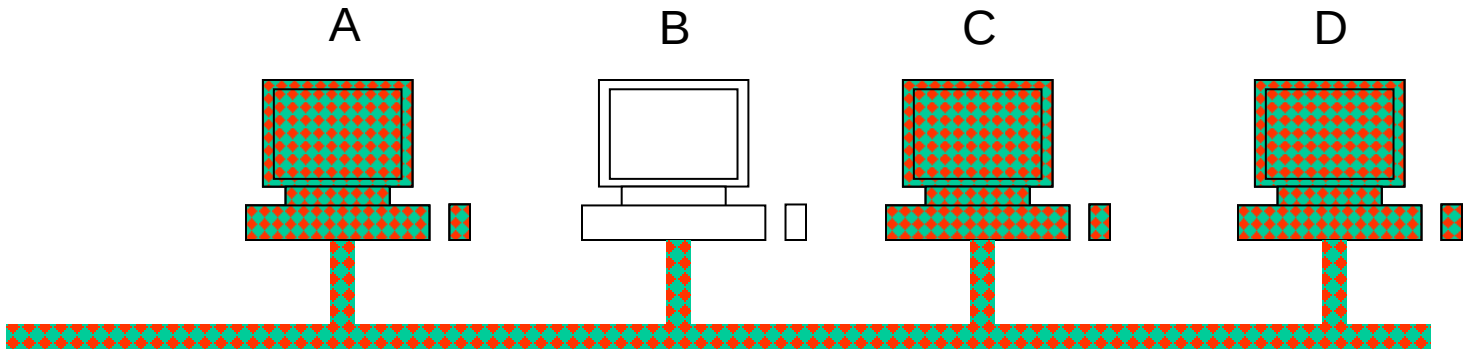
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CSMA/CD (2)



1. Station A wants to trans
2. Station A starts t
3. Station C receives the beginning of
4. Station D wants to transmit to stati diu
5. There is a collision.

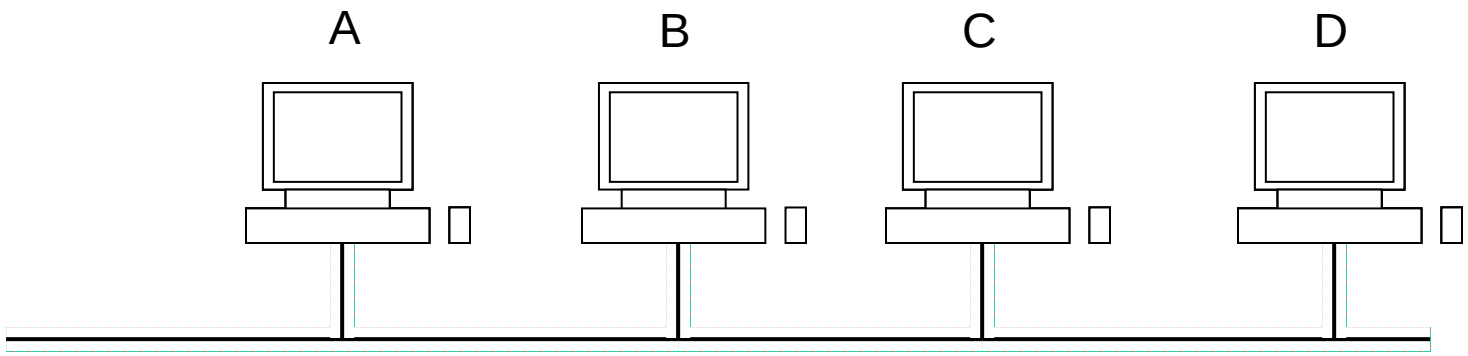
CSMA/CD (3)



1. Station A wants to transmit to station C and senses a free medium.
2. Station A starts the transmission.
3. Station C receives the beginning of the message.
4. Station D wants to transmit to station B and senses a free medium.
5. There is a collision.

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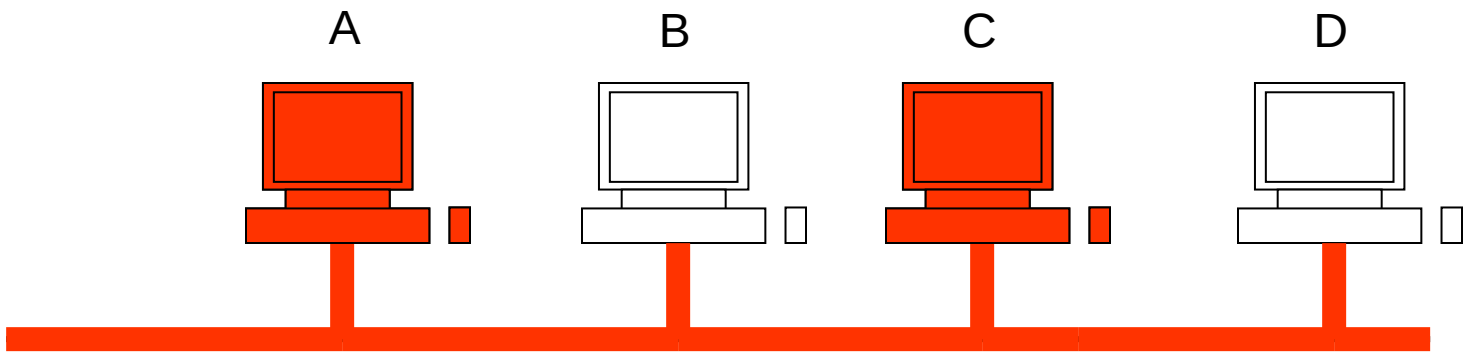
CSMA/CD (4)



1. Station A wants to transmit to station C and senses a free medium.
2. Station A starts the transmission.
3. Station C receives the beginning of the message.
4. Station D wants to transmit to station B and senses a free medium.
5. There is a collision.
6. A and D interrupt their transmissions.

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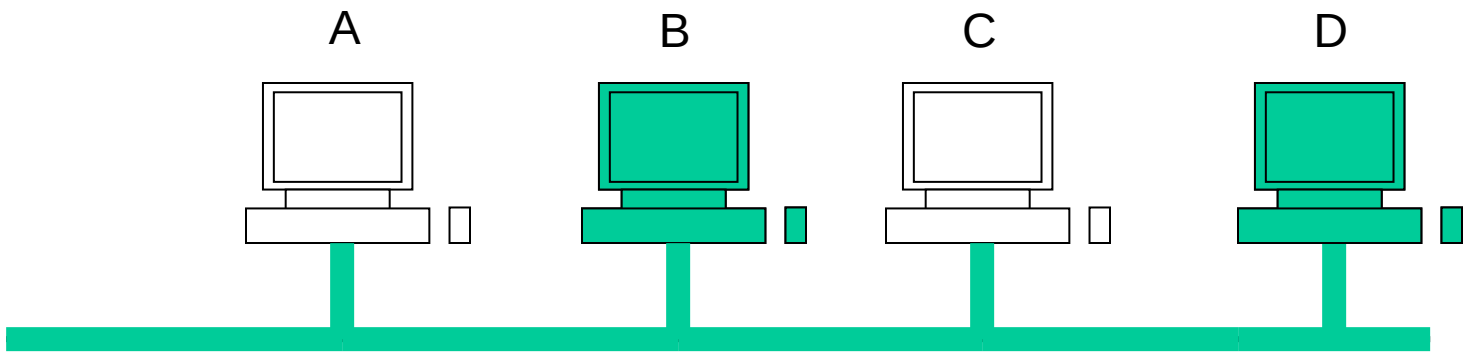
CSMA/CD (5)



1. Station A wants to transmit to station C and senses a free medium.
2. Station A starts the transmission.
3. Station C receives the beginning of the message.
4. Station D wants to transmit to station B and senses a free medium.
5. There is a collision.
6. A and D interrupt their transmissions.
7. A tries again after a random backoff time. This time the station "wins".

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CSMA/CD (6)



1. Station A wants to transmit to station C and senses a free medium.
2. Station A starts the transmission.
3. Station C receives the beginning of the message.
4. Station D wants to transmit to station B and senses a free medium.
5. There is a collision.
6. A and D interrupt their transmissions.
7. A tries again after a random backoff time. This time the station "wins".
8. D tries again after a random backoff time. This time the station gains access.

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"Truncated Binary Exponential Backoff"

In case of light load

stations almost always have **immediate access** to the shared medium.

With higher load

the collision probability increases:

It becomes more and more likely that several stations start a transmission (almost) simultaneously.

Obviously, a **mechanism for load reduction / traffic shaping** is required:

Light load:	choose a **small backoff time**
Heavy load:	choose a **large backoff time**

In real life, the so-called **"Truncated Binary Exponential Backoff"*** is used.
With this mechanism, the current load is estimated from the number of collisions the current message was already involved in.

Many collisions already:	**large backoff time**
Small # of collisions so far:	**small backoff time**

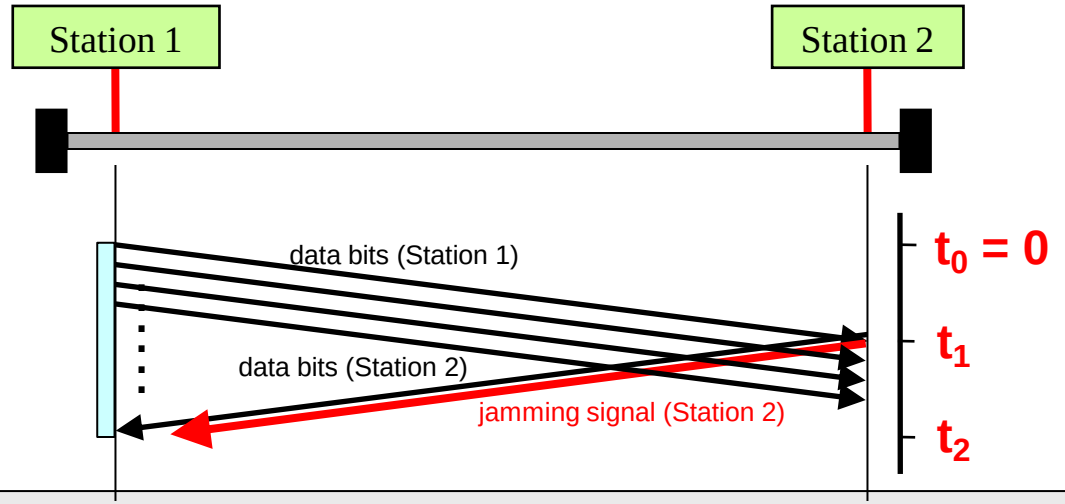
***A similar mechanism is used in 802.11 WLAN**

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"Truncated Binary Exponential Backoff" (2)

Minimum frame length 64 bytes:
- A station needs to send for at least one slot time
- To still detect collision in the worst case scenario of max. distance between stations

Station 1 also starts to send a jamming signal after t_2



Let t_0 be the time when station 1 starts transmitting

t_1 be the time when station 2 starts transmitting (sensing free medium), shortly after that its transmission collides with the transmission of station 1, station 2 starts to send a jamming signal

t_2 be the time when station 1 detects the collision

Then: $t_1 \leq$ **end-end-delay**; $t_2 \leq$ **2 • end-end-delay**

(so-called "Slot Time")

k^{th} re-transmission ($k \leq 10$): Randomly choose a backoff time as r * slot time with $0 \leq r < 2^k$ (r integer, Ethernet slot time: 51.2 µs)

k^{th} re-transmission ($10 < k \leq 16$): same as above, but $0 \leq r < 2^{10}$

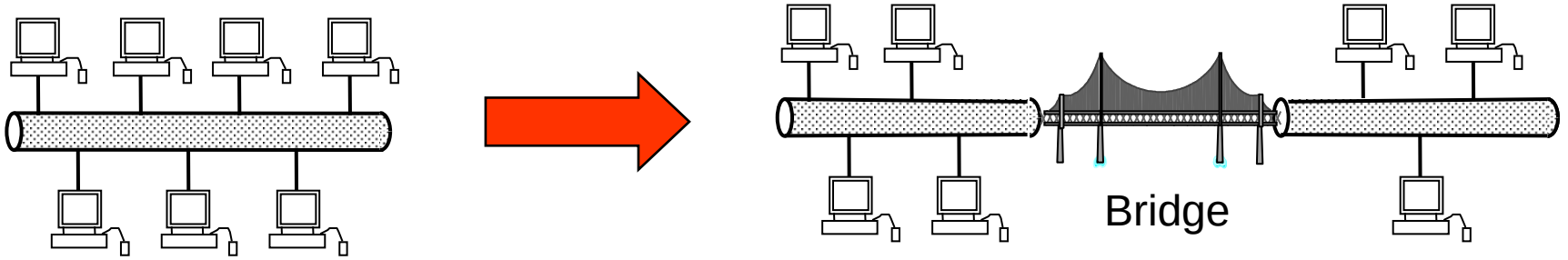
If the transmission still fails: no further attempts.

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Migration paths of Ethernet and IEEE 802.3

What can you do if the demand exceeds the available bandwidth ?

a) Split the network into several subnetworks:



Load separation is only achieved with an appropriate communication pattern.

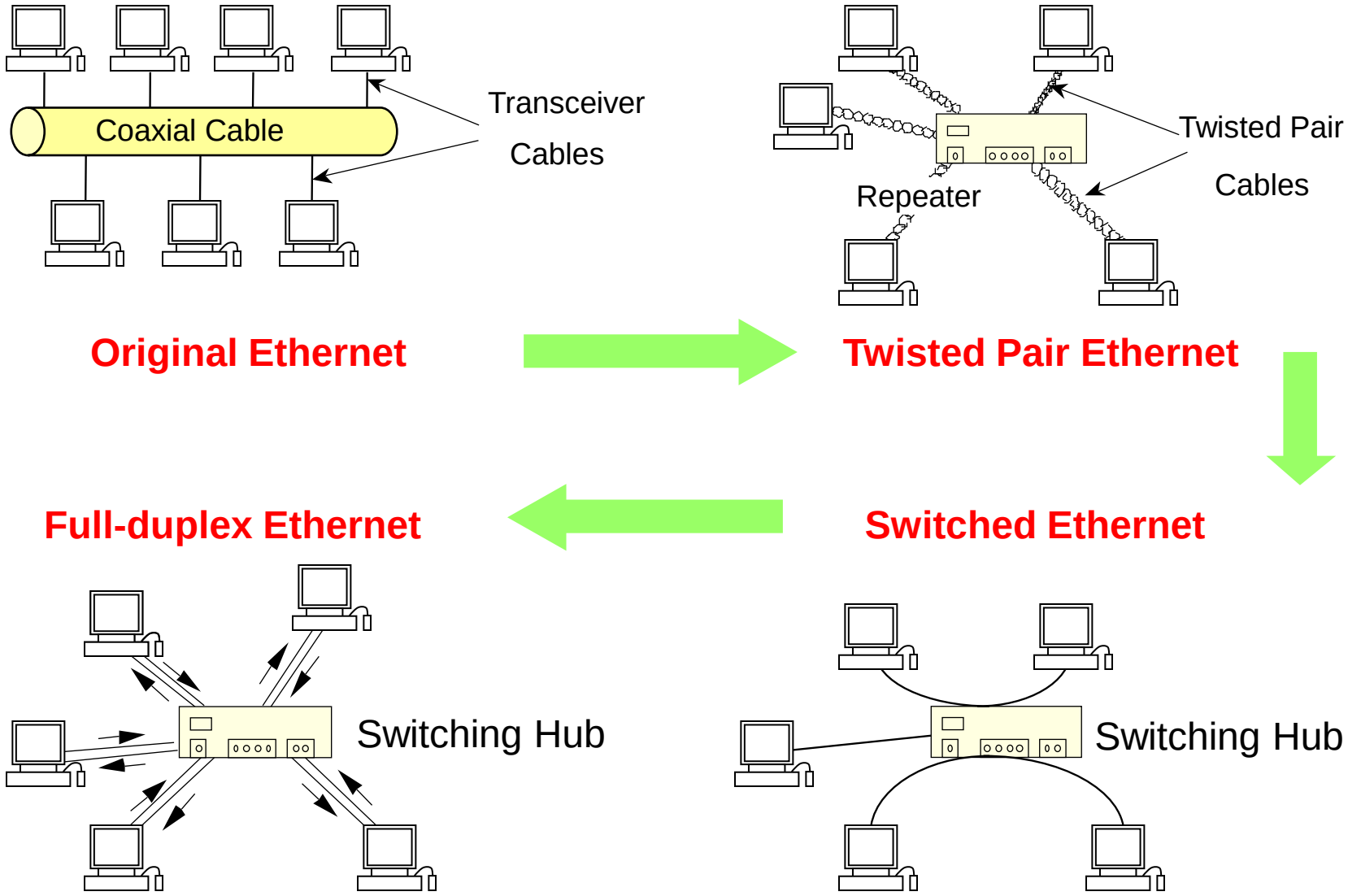
b) Use the faster but smaller ..: 100 Mbit/s-Ethernet

c) Use multiport bridges, i.e. "switches": Switched Ethernet

d) Do not use CS, MA and CD: Full-duplex Ethernet

The service access points used by higher layer protocols remain the same !

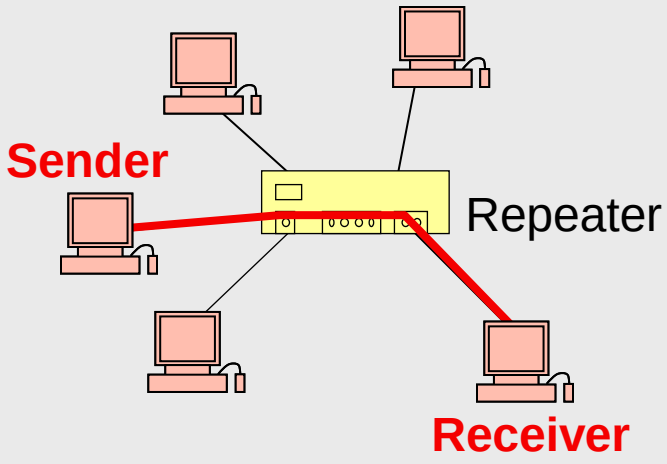
"Ethernet": Yesterday - today - tomorrow



Switched Ethernet

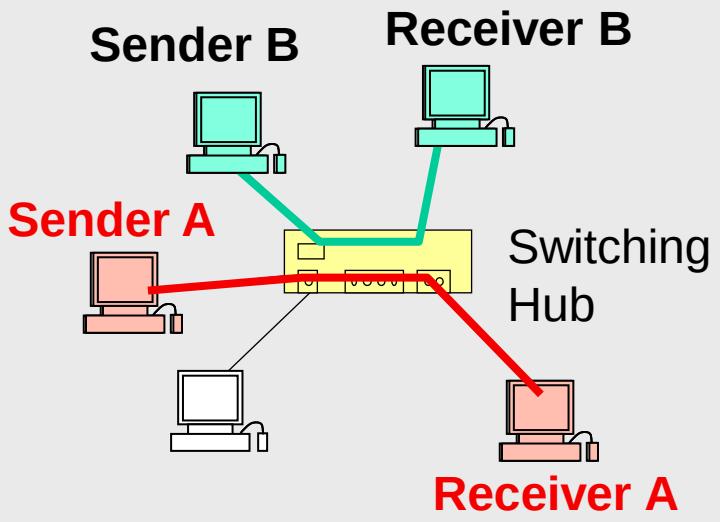
Ethernet (classic):

- End system “sees” all messages in the network.
- At any point in time, at most one end system may transmit data.
- CSMA/CD detects and resolves collisions.
- For each transmission there is quite a large number of potential competitors.



Switched Ethernet:

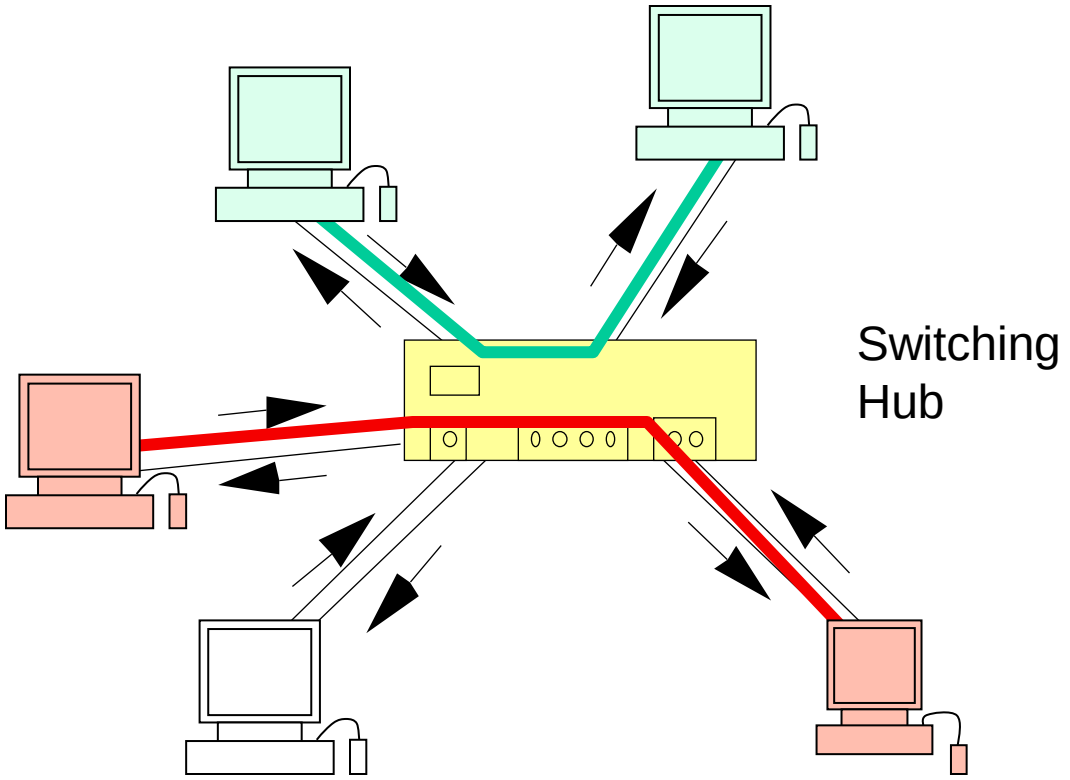
- Messages are forwarded to the receiver only.
- Several end systems may transmit simultaneously.
- CSMA/CD detects and resolves collisions.
- For each transmission, there is only 1 competitor (Hub or end system).



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Full-duplex Ethernet

End systems and switches are connected by full-duplex links.



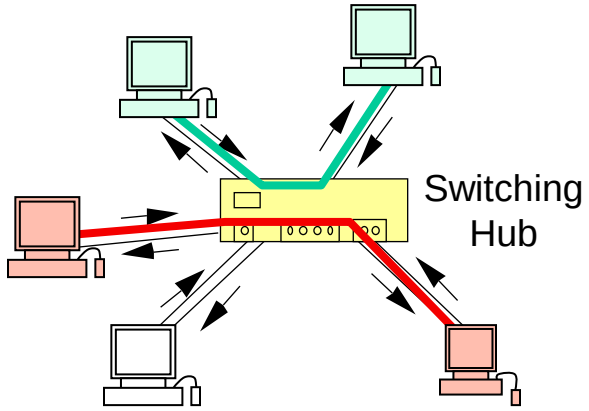
All links are used bi-directionally.

There are no collisions.

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Full-duplex Ethernet (IEEE 802.3x)

End systems and switches are connected by full-duplex links.

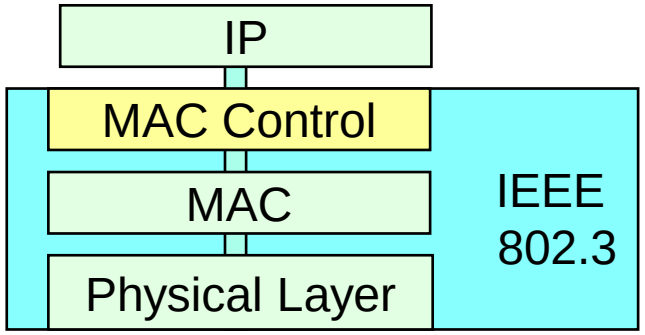


All links are used bi-directionally.

There are no collisions.

With the collisions we also lose the "natural" flow control.

The standard IEEE 802.3x adds a new sublayer ("MAC Control") taking care of flow control.



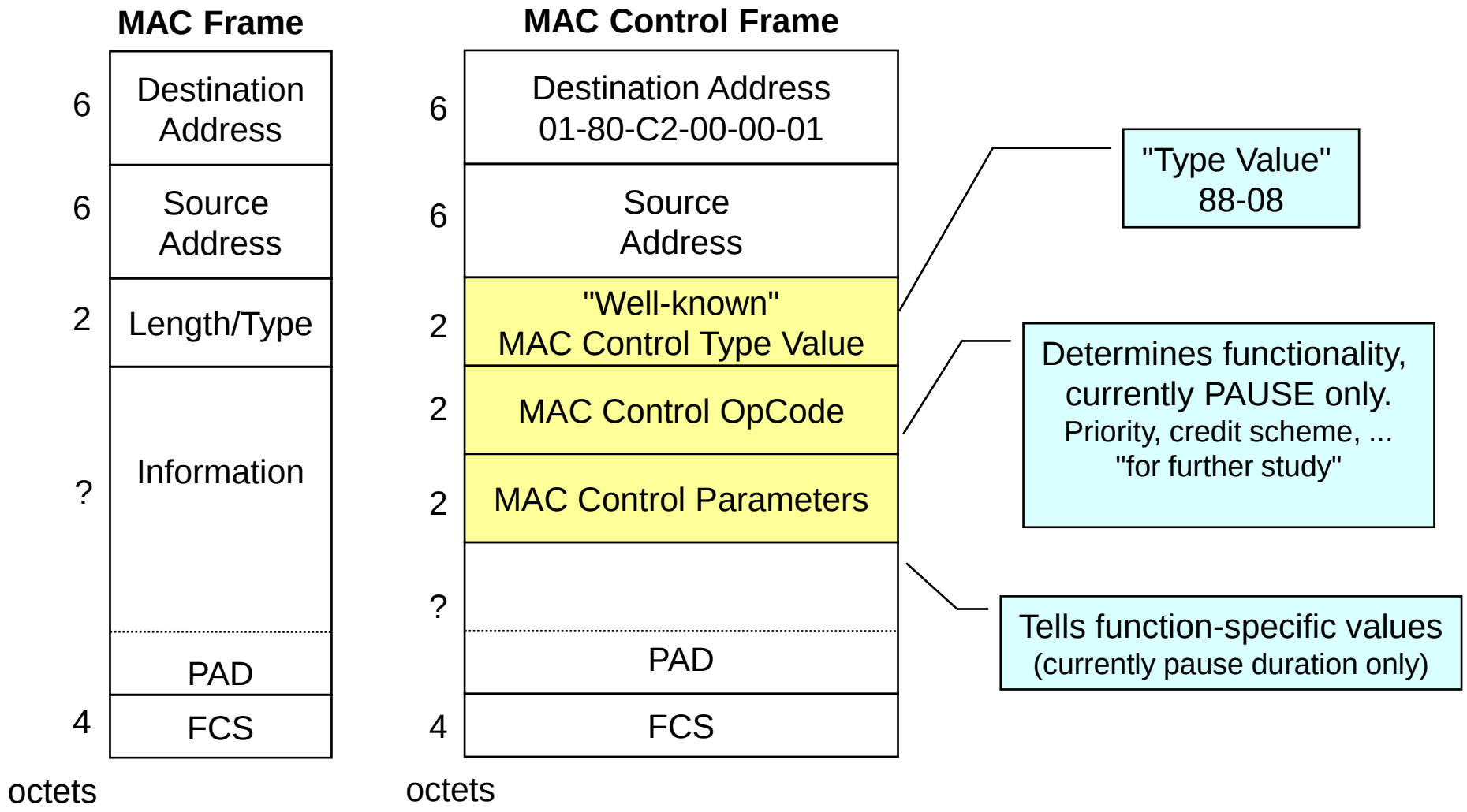
In case of overload, the receiver tells the sender what to do :

"MAC Control Frames" (in the opposite direction!)

MAC Control Receive State Machine

MAC Control Transmit State Machine

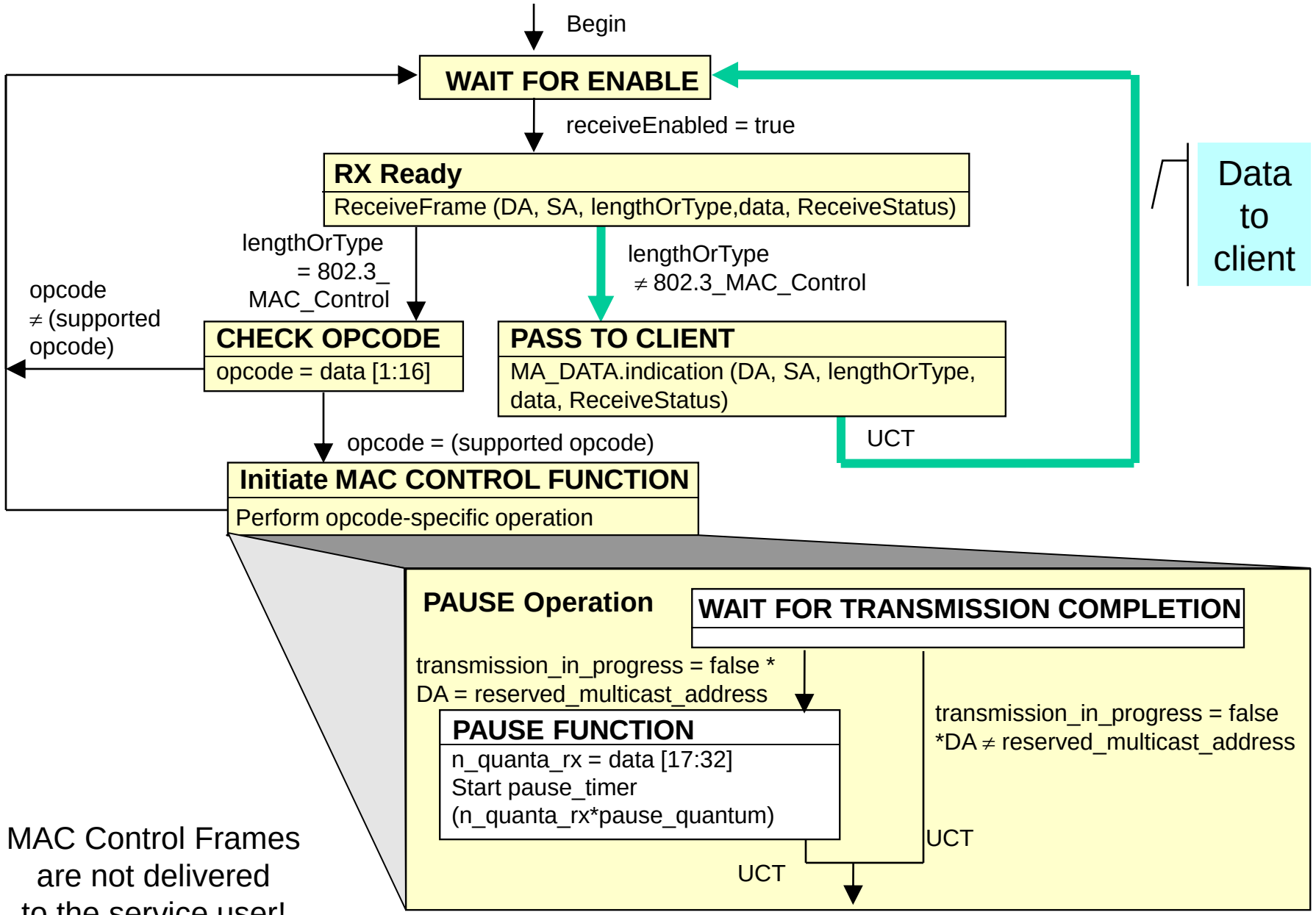
MAC Control Frames (Full-duplex Ethernet)



MAC-Control Frames are generated and "consumed" in the MAC control sublayer.
In general, they do not reach the MAC service user.

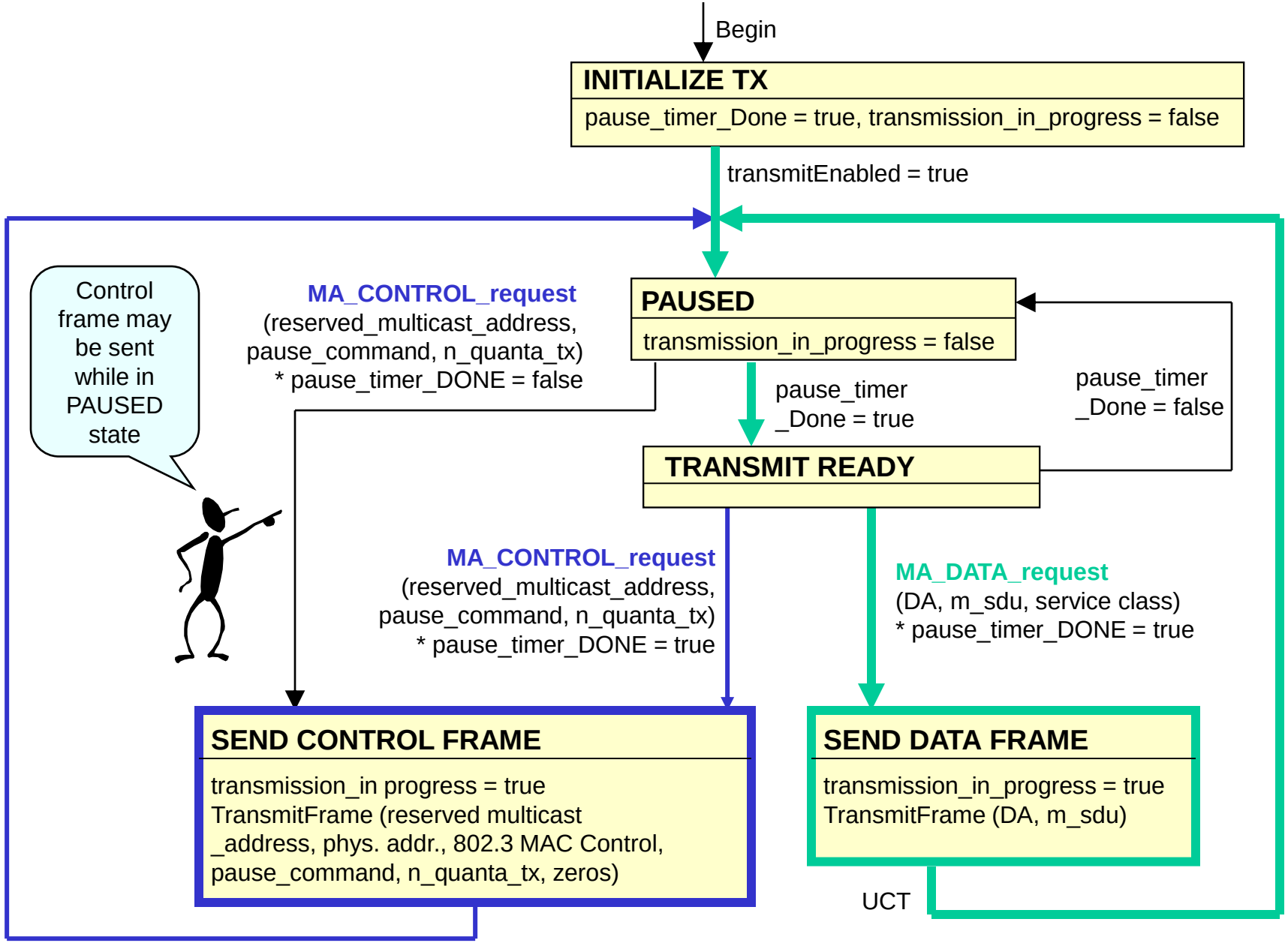
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MAC Control Receive State Machine (Pause Operation)



MAC Control Frames are not delivered to the service user!

MAC Control Transmit State Machine (Pause Operation)



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Gigabit Ethernet (IEEE 802.3z)

"Gigabit Ethernet employs
- the same Carrier Sense Multiple Access with Collision Detection protocol,
- the same frame format and
- the same frame size as its predecessors."

"For the vast majority of network users, the existing network investment can be extended to gigabit speeds at reasonable initial cost
- without the need to re-educate their support staff and users, and
- without the need to invest in additional protocol stacks or middleware. The result is low cost of ownership for users."

"Gigabit Ethernet is Ethernet, only faster."

From the White Paper: "Gigabit Ethernet - Accelerating the standard for speed"

Source: Gigabit Ethernet Alliance, 1997 http://www.gigabit-ethernet.org/

originally re-directed to: http://www.10gea.org/Tech-whitepapers.htm

but both no longer working as originally!

Changes when compared to original Ethernet

1. New minimum frame length when using CSMA/CD:

Original Ethernet:	Minimum 64 byte (cf. slide 74)
Gigabit-Ethernet:	Minimum 512 byte (by "Carrier Extension")
Reason:	Makes sure that transmission time > signal propagation time

2. New slot time (used by backoff) when using CSMA/CD

Original Ethernet:	64 byte times (cf. slide 74)
Gigabit-Ethernet:	512 byte times
Reason:	Change to minimum frame length

3. Option to avoid CS, MA and CD

Original Ethernet:	Half-duplex transmission with CSMA/CD
Gigabit Ethernet:	Full-duplex transmission (IEEE 802.3x) possible, if medium > 100 m mandatory (then: No change to min. frame length and slot time)
Reason:	Significantly better efficiency and - basically - no restrictions to medium length (otherwise max. 100 m)

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3.6.4. WLAN ... in a Nutshell

Wireless LAN (WLAN) is also called “Wireless Ethernet”.

In fact, Ethernet and WLAN have several characteristics in common:

- **Shared medium:** The wireless medium is shared by all stations in the area.
- **Risk of collisions:** Collisions may happen.

However, there are important differences:

- **Collision Detection:** In WLAN, collisions are hard to detect
  (The transmitted signal is much stronger than the received signal)
- **Network Formation:** Who belongs to the network?
  (In Ethernet, you can check the cables. In WLAN, you need identifiers)

3.6.4.1. Wireless Links

3.6.4.2. Basic Building Blocks of Wireless LAN

3.6.4.3. The Distribution System

3.6.4.4. Wireless Medium Access Methods

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3.6.4.1 Wireless Links

As in wired communication, a transmission in wireless systems is between neighbouring stations on the same **physical link.** However, a link in wireless systems behaves **completely different** than in wired systems.

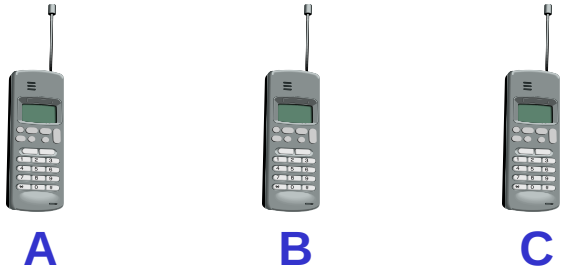
Wired	Wireless
low bit error rate	high bit error rate (10^{-1} to 10^{-2})
constant propagation characteristics	(rapidly) changing propagation characteristics
constant quality of the link	short periods with no communication (deep fades)
clear separation of links by cables	unclear boundaries of a link

!!! While a link in a wired channel is deterministic, **a link in wireless communications has probabilistic characteristics**

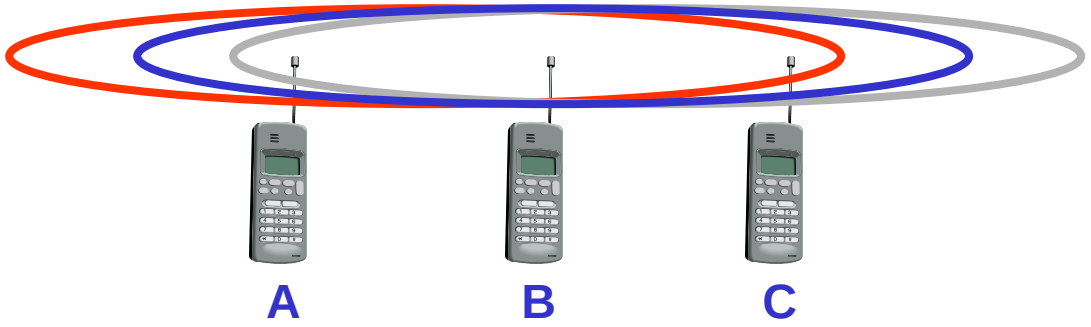
Note: This has **consequences** on the networking mechanisms:
Further **protection and error recovery** mechanisms have to be introduced

Wireless Links (II)

Opposed to wired communication, links have no clear separation in wireless communication. Whether stations A, B, and C are on the same link depends on the communication ranges of the stations.

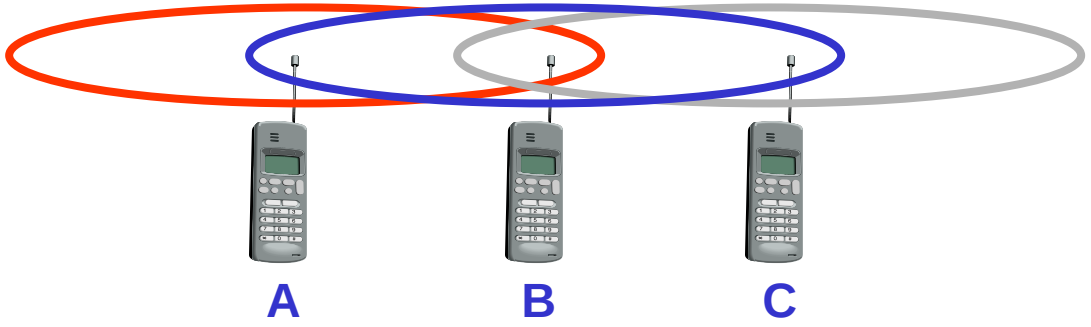


For sufficient transmission ranges, all three devices form **a single link**.



Wireless Links (II)

For smaller transmission ranges, the devices form **different links**, one containing stations A and B, the other containing stations B and C.



This means:

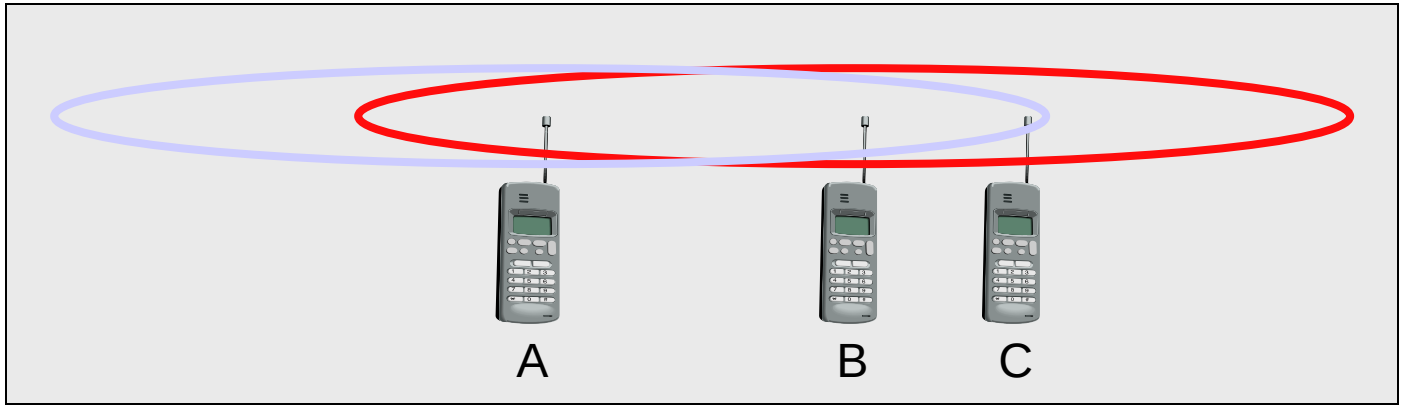
- C might **not even be aware** of A's presence (A is hidden to C)
- A might **interfere** on BC's link (without even being aware of this)
- The **available bandwidth on a link** does not only depend on the load on that link, but also on the load on different links. Ultimately, the available bandwidth on a link cannot be known from information from the local link alone.

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Near and far terminals

Terminals A and B send, C receives

- signal **strength decreases** according to the specific path loss
- the **signal** of terminal B therefore **drowns out** A's signal
- **C cannot receive** A



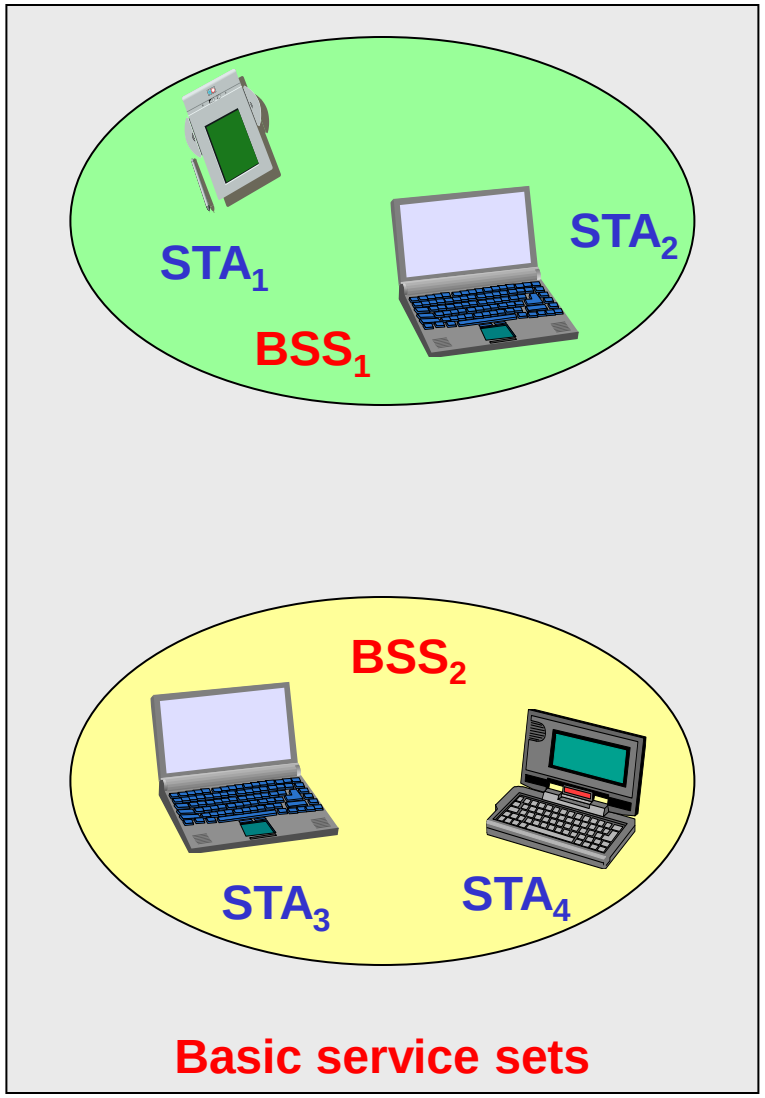
Source: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003

If C for example was an arbiter for sending rights, terminal B would drown out terminal A already on the physical layer.

Therefore, many wireless systems provide means for **power control.**

3.6.4.2. Basic Building Blocks of Wireless LAN

The **basic building block** of an IEEE 802.11 LAN is called **basic service set (BSS)**.



Basic service sets

Station (STA):

- **Any device** that contains an **IEEE 802.11 conformant medium access control (MAC) and physical layer (PHY)** interface to the wireless medium (WM).

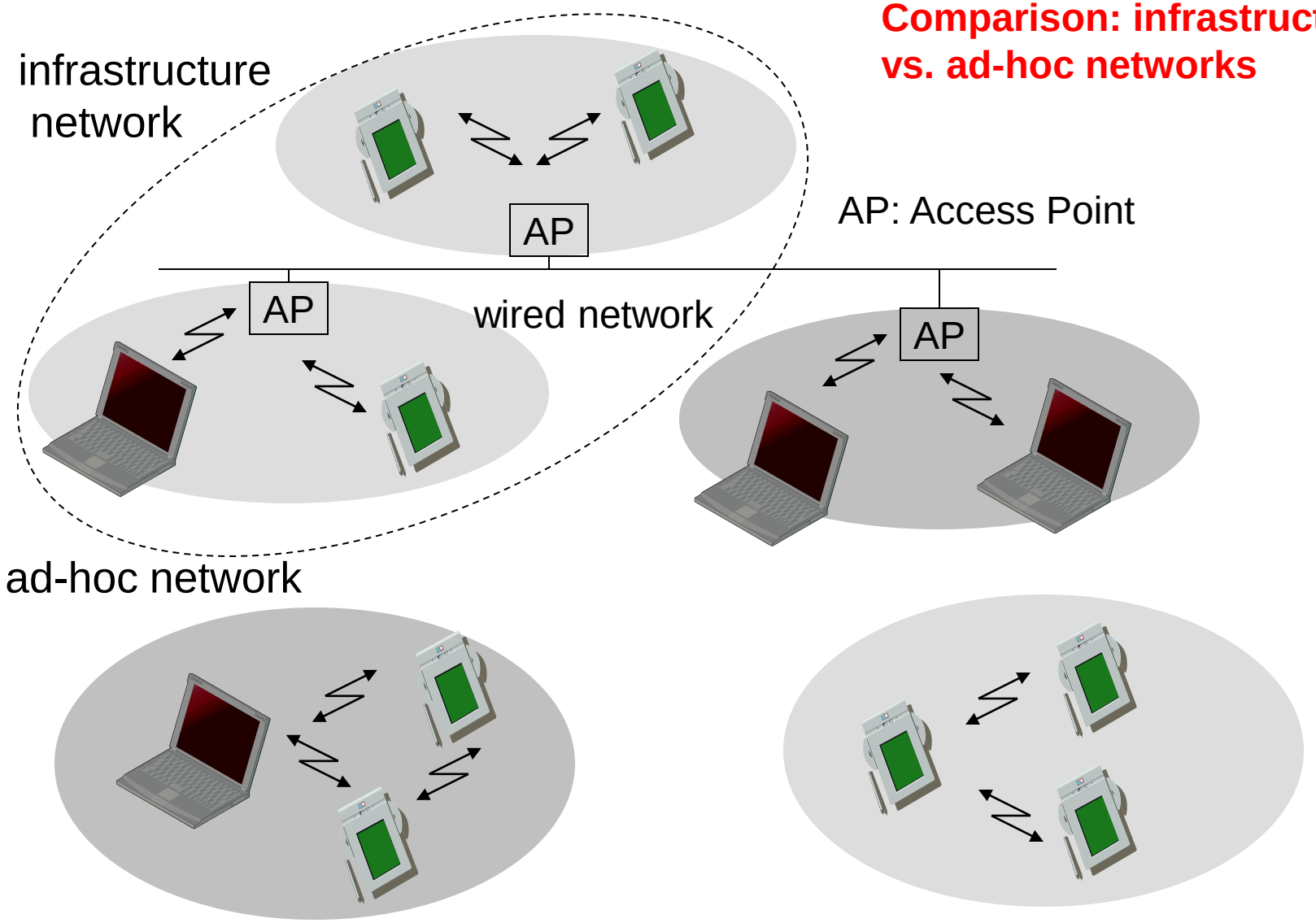
Basic service set:

- It is useful to think of the ovals used to depict a BSS as the **coverage area within which the member stations of the BSS may remain in communication.**

- The **concept of area**, while not precise, is **often good enough**.
- If a station **moves out** of its BSS, it can **no longer directly communicate with other members of the BSS.**

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Components & services of the IEEE 802.11 architecture

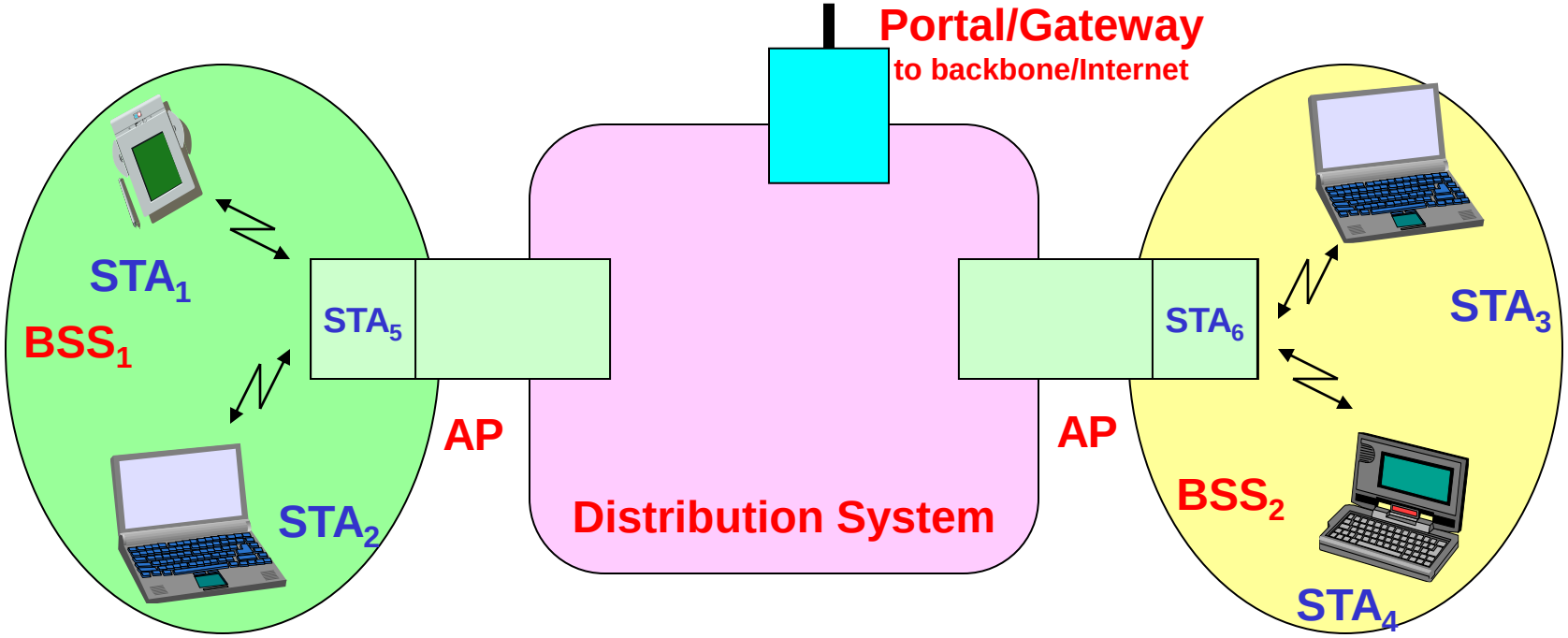


Comparison: infrastructure vs. ad-hoc networks

Source: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003

3.6.4.3. The Distribution System

In some networks, **coverage beyond the direct station-to-station distance** is required.
BSSs may be interconnected by a **“distribution system” (DS)**.



The DS (usually a **conventional LAN** with or without connection to the Internet) provides the logical services necessary to handle **address to destination mapping**.

Data (packets) move between a BSS and the DS via an **access point (AP)**.
Note that **all APs are also STAs**; thus they are **addressable entities**.

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3.6.4.4. Wireless Medium Access Methods

Fundamental access method:

- **“Distributed Coordination Function”** (**DCF**),
- **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA).**
- is implemented **in all STAs** (both IBSS and infrastructure network configurations)

Collisions are avoided by **random backoff times after busy medium conditions**.

All directed traffic uses immediate **positive acknowledgment** (ACK frame).

Retransmission is scheduled by the sender if no ACK is received.

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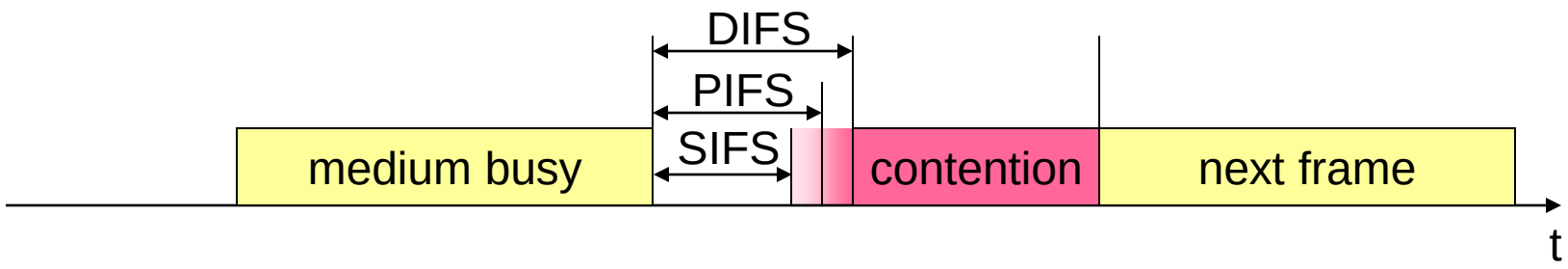
The distributed coordination function (DCF)

The DCF allows for automatic medium sharing through the use of
• **CSMA/CA** with a **random backoff time following a busy medium condition**.

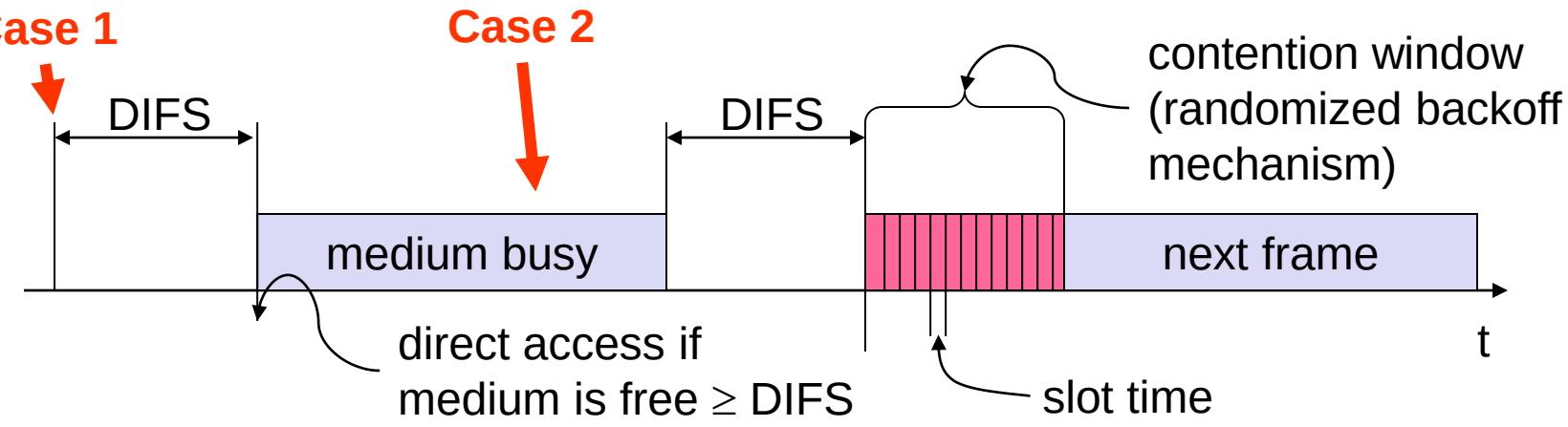
All directed traffic uses immediate **positive acknowledgment** (ACK frame) where retransmission is scheduled by the sender if no ACK is received.

IEEE 802.11 defines **access priorities through different inter frame spaces:**

- **SIFS** (Short Inter Frame Spacing)
  - **highest priority**, for ACK, CTS, polling response
- **PIFS** (PCF IFS)
  - **medium priority**, for time-bounded service using PCF (point coordination function)
- **DIFS** (DCF, Distributed Coordination Function IFS)
  - **lowest priority**, for asynchronous data service



CSMA/CA, mandatory (2 cases!)

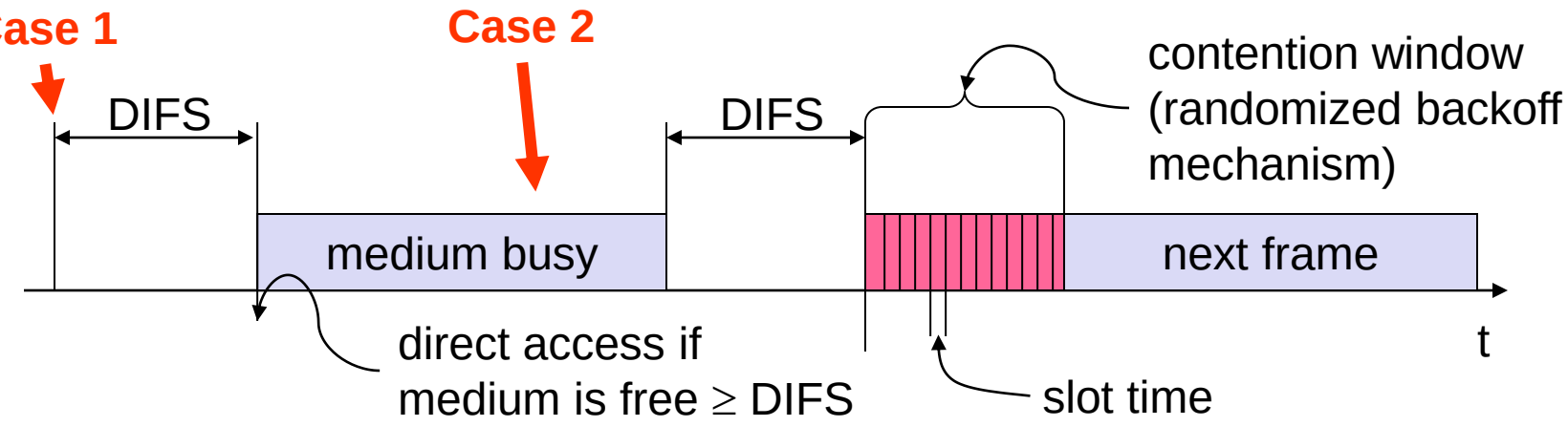


Source: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003

- when ready: start **sensing the medium**
- **Case 1: if the medium is free** for the duration of an Inter-Frame Space (IFS), the station can **start sending** (IFS depends on service type)
- **Case 2: if the medium is busy**,
  - the station has to **wait for a free IFS**,
  - then the station must **ADDITIONALLY WAIT for a random backoff time**
    (collision avoidance, multiple of slot-time)
- **if another station occupies the medium during the backoff time** of the station, the **backoff timer stops** (fairness)
- IEEE 802.11 uses **exponential backoff**: The **contention window doubles** with each collision (cf. exponential backoff of CSMA/CD).

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CSMA/CA, mandatory (2 cases!)



Source: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003

- when ready: start **sensing the medium**
- **Case 1: if the medium is free** for the duration of an Inter-Frame Space (IFS), the station can **start sending** (IFS depends on service type)
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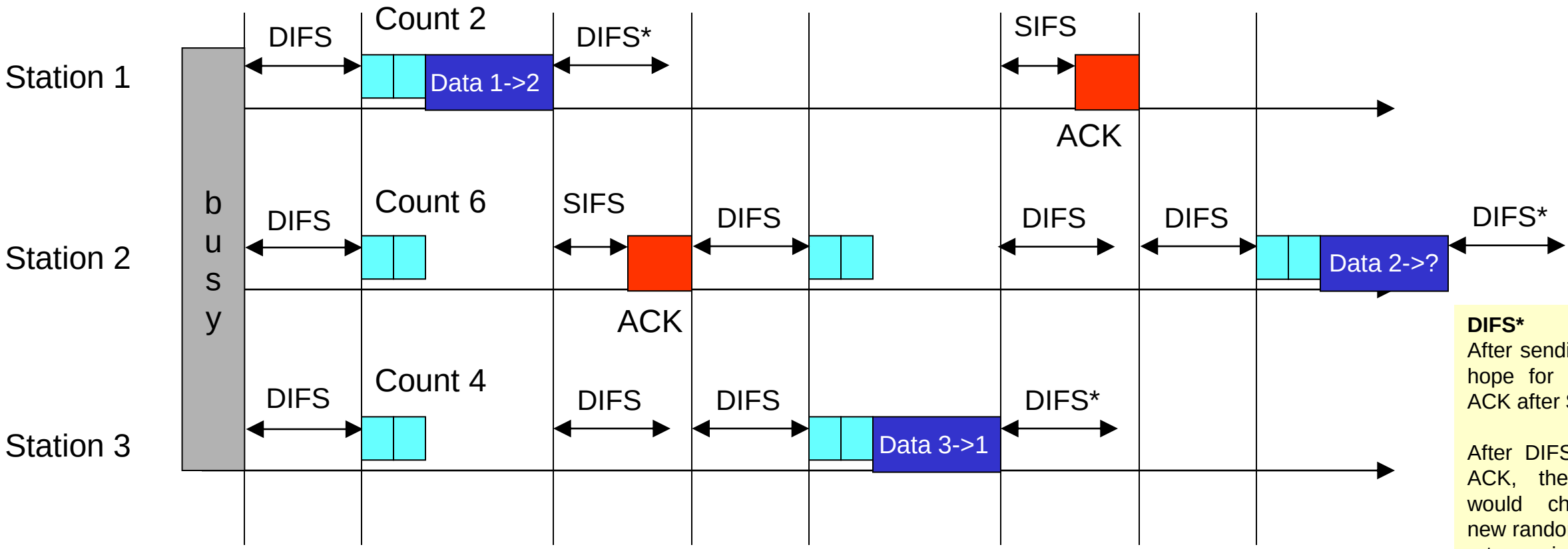
If another station starts to send before the IFS has elapsed:

- do NOT start to send!
- Wait until the medium is free again

- Case 1 then turns to Case 2 (!)
- Case 2 restart with the chosen random backoff or with the residual backoff time (**see examples on next slides**)

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CSMA/CA MAC: Example 3 stations (each wants to send a data frame)



DIFS*
After sending Data, hope for incoming ACK after SIFS.

After DIFS without ACK, the station would choose a new random BO for retransmission!

CSMA: Carrier Sense, Multiple Access: busy medium => no access

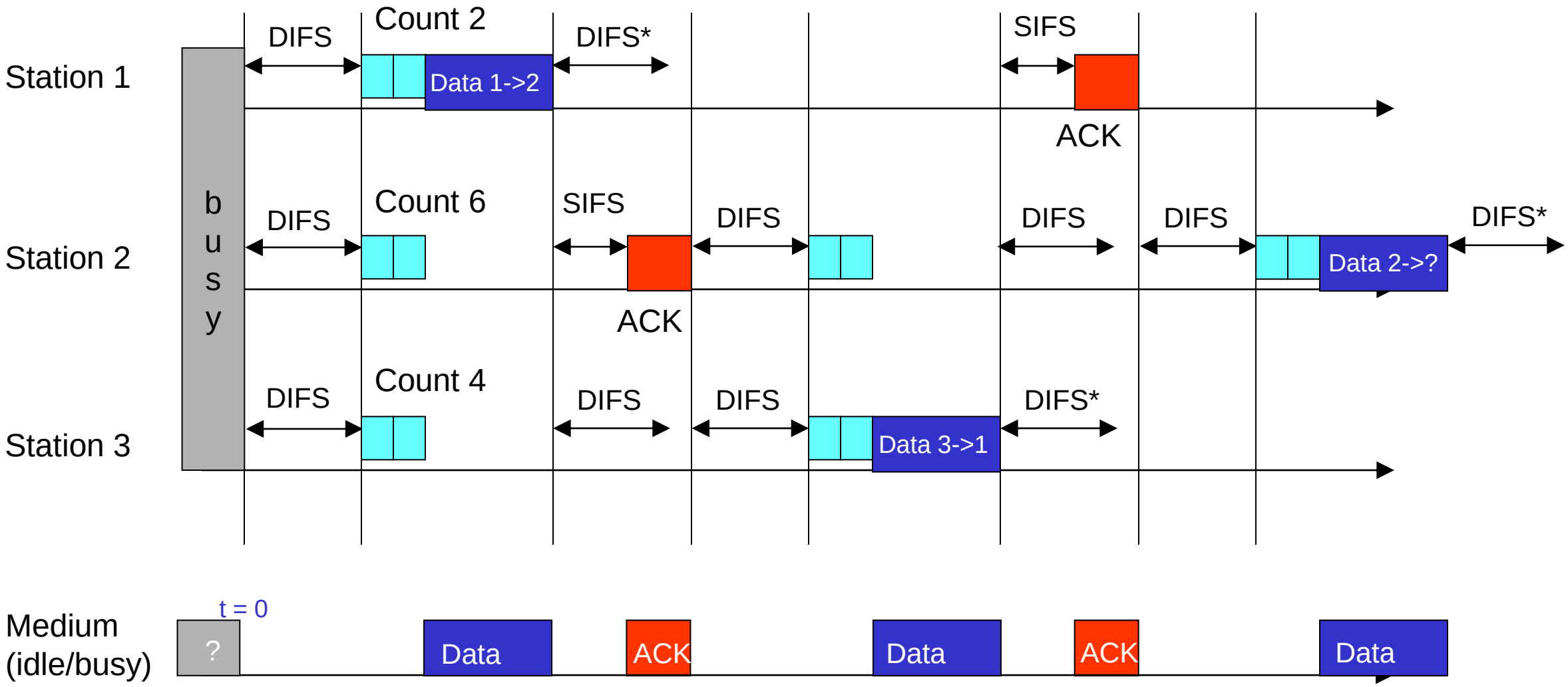
CA: Collision Avoidance: select random backoff-time (multiple of constant slot-time)

t = 0, medium gets idle

Collision Detection is not possible (wireless!)
=> receiver sends ACKnowledgement with highest priority (SIFS)
lack of ACK triggers retransmission

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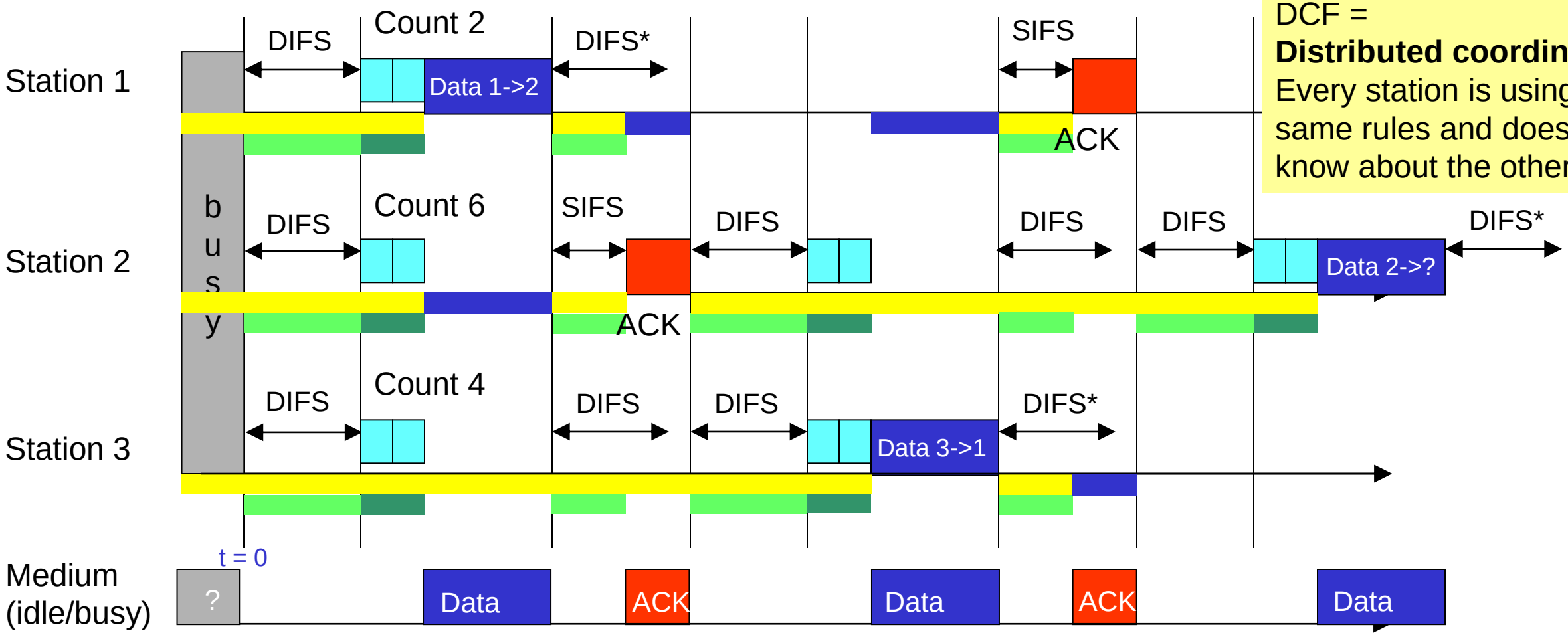
CSMA/CA MAC: Example 3 stations (legend + additional hints)



Think about the (raw) network/medium utilisation!
(IFS waiting time and counting random backoff are “wasted” time!)

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CSMA/CA MAC: Example 3 stations (legend + additional hints)



DCF = **Distributed coordination**! Every station is using the same rules and does not know about the others!

Color legend

(Yellow) Station doing **carrier sense** (CS)

(Light green) Station **waiting for IFS** (either DIFS or SIFS)

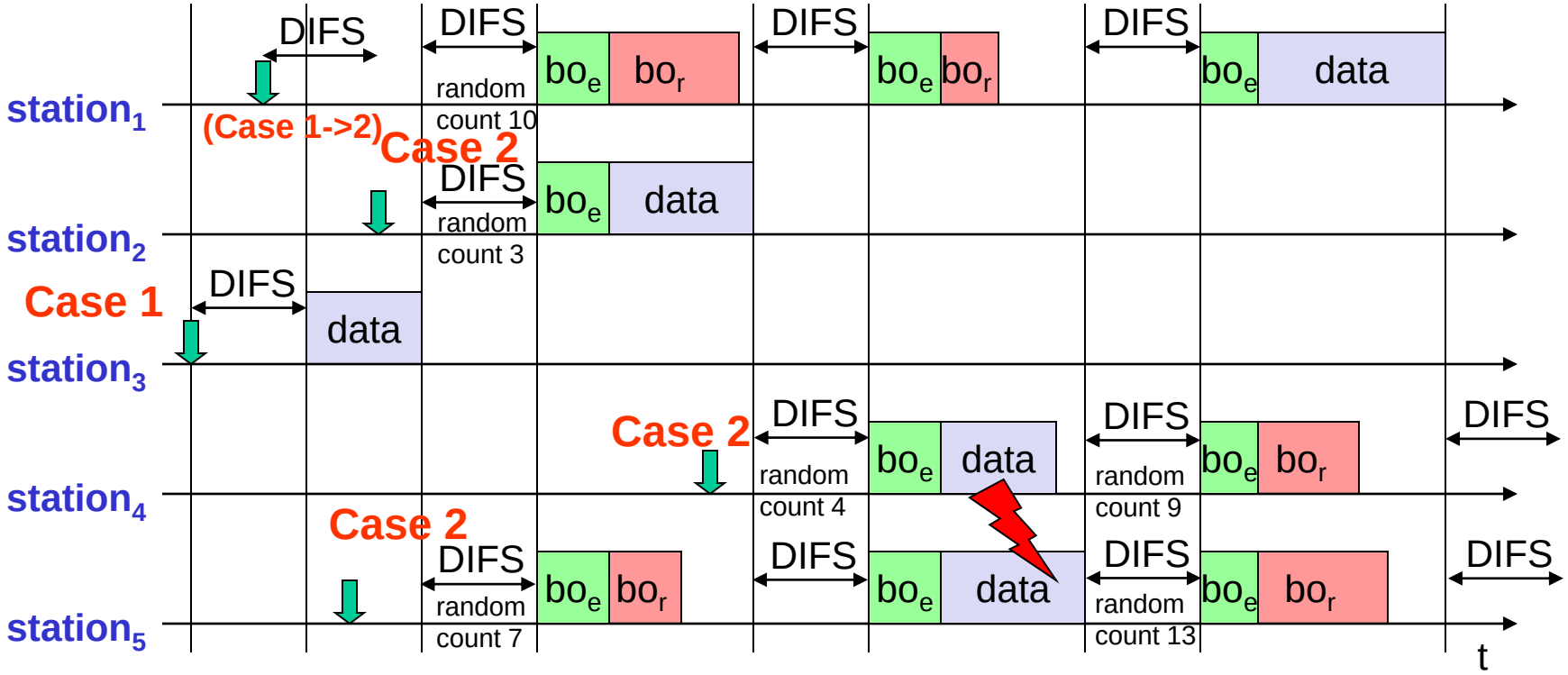
(Dark green) Station **counting down random backoff**

(Blue) **Station receiving** Data or ACK

Basic idea: 5 stations competing for access

(example for collision: station 4 joins late, gets into collision with station 5)

Figure slightly modified from original: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003



Station 4 and 5 are expecting ACK!

No ACK => **collision!**

Start retransmit A.S.A.P. with ***NEW* random backoff time**! (exponential backoff)

busy — medium not idle (data, ACK etc.)

bo_e — elapsed backoff time

packet arrival at MAC

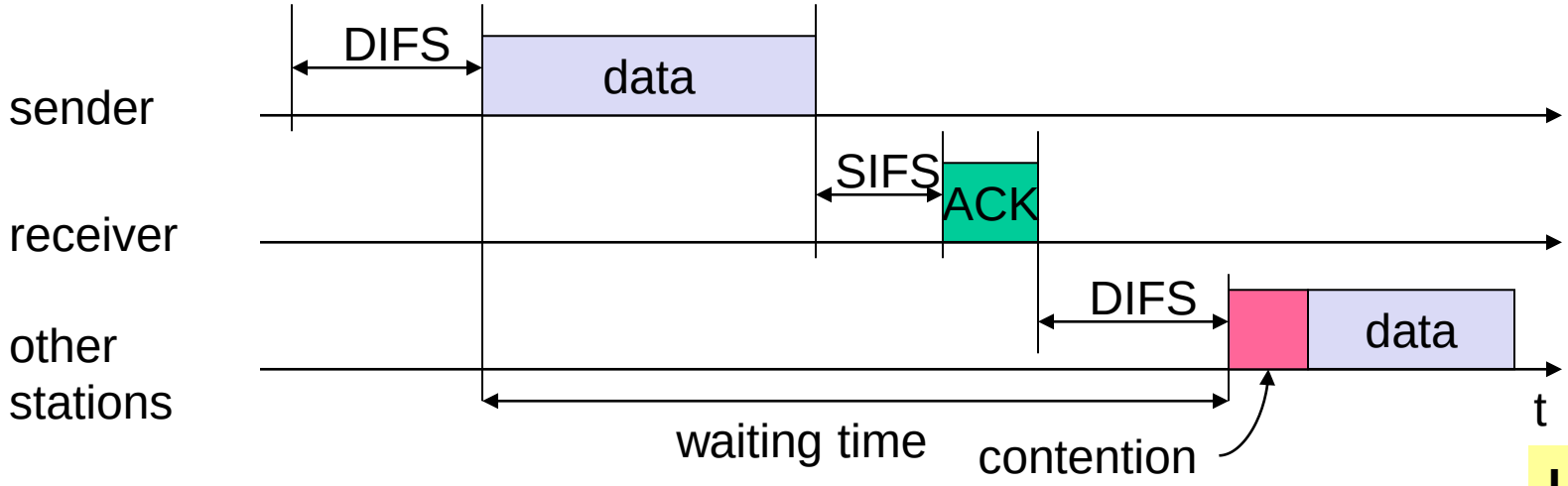
bo_r — residual backoff time

Details with SIFS + Acknowledgements (ACK) omitted for sake of readability. Every successful transmission will be followed by SIFS + transmission of ACK (cf. Slides 97 or 101).

Acknowledgements

Instead of Collision Detection, **IEEE 802.11 uses ACKs**:

- station has to **wait for DIFS** before sending data
- **receivers acknowledge at once** (after waiting for SIFS) if the packet was received correctly (CRC)
- **automatic retransmission** of data packets in case of transmission errors



Source: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003

Lost ACK!
Cf. ch. 3.4 alternating bit sequence numbering!

Duplicate frames (lost ACK) shall be filtered out within the destination MAC.

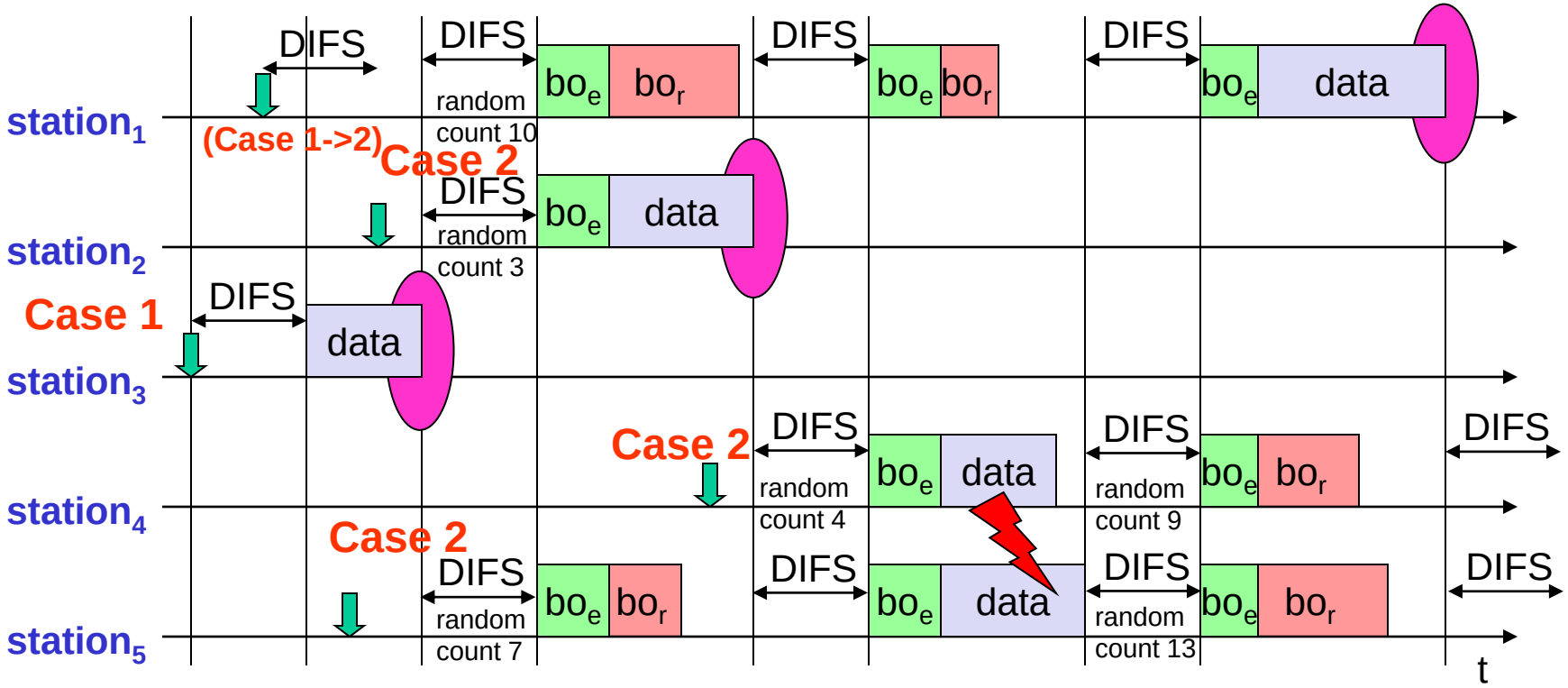
This is facilitated through a **Sequence Control field** (sequence number + fragment number) within data and management frames.

The sequence number is generated by the transmitting STA as an **incrementing sequence of integers.**

Basic idea: 5 stations competing for access

(example for collision: station 4 joins late, gets into collision with station 5)

Figure slightly modified from original: Jochen Schiller: "Mobile Communications", Addison-Wesley, 2003



Station 4 and 5 are expecting ACK!

No ACK => **collision!**

Start retransmit A.S.A.P. with ***NEW* random backoff time**! (exponential backoff)

busy — medium not idle (data, ACK etc.)

bo_e — elapsed backoff time

packet arrival at MAC

bo_r — residual backoff time

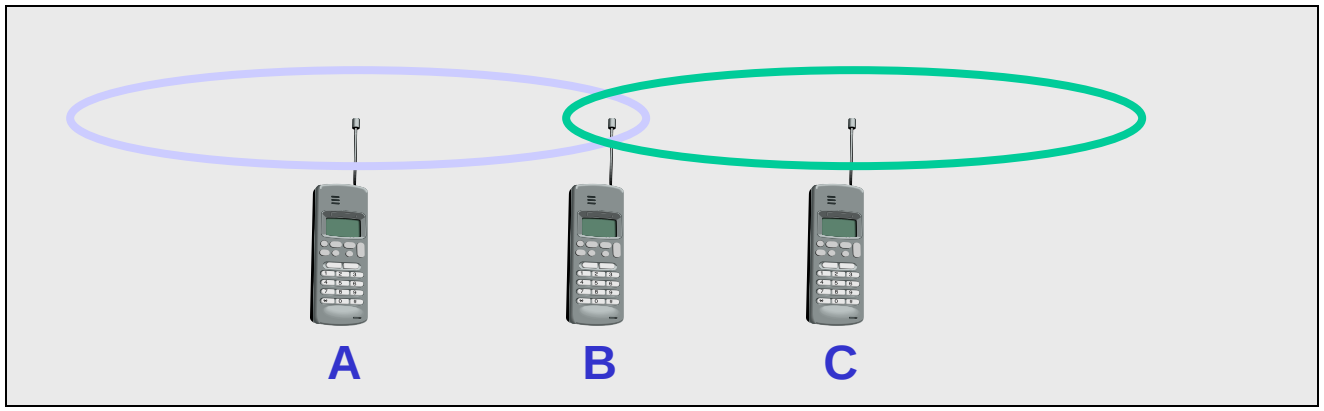
Details with SIFS + Acknowledgements (ACK) omitted for sake of readability.
Every successful transmission will be followed by SIFS + transmission of ACK (cf. Slides 97 or 101).

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Hidden terminals

Hidden terminals

- A sends to B, C cannot receive A
- C wants to send to B, C senses a “free” medium (CS fails)
- collision at B, A cannot receive the collision (CD fails)
- A is “hidden” for C



Source: Jochen Schiller: “Mobile Communications“, Addison-Wesley, 2003

Comparison Ethernet - WLAN

	Ethernet	WLAN – "Wireless Ethernet"
CS	sense medium before sending	sense medium before sending
MA	all stations share the medium (cable)	all stations share the medium (radio frequency)
collisions (CD vs. CA)	collision detection (CD): physically detect collision, stop transmission, retry after random backoff	collision avoidance (CA): when medium getting free, wait for **DIFS** + random contention period/backoff collision is detected by ACKnowledgement mechanism (using **SIFS** for ACK)
operation without collisions	full-duplex Ethernet	PCF – Point Coordination Function (uses **PIFS** = PCF IFS)